INCH-POUND

MIL-M-28787/5C
7 September 1994
SUPERSEDING
MIL-M-28787/5B(EC)
31 January 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIPLEXER, DIGITAL, KEY CODE BDL

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

*The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

* 1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic multiplexer module.

* 1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603A, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* 3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

* 3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

* 3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.115 failures per million operating hours for class I or 0.168 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389.

* 3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

* 3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

* 3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

* 3.6.1 Key code. The module key code shall be BDL.

* 3.6.2 Size. The module size shall be 1A.

* 3.6.3 Weight. The module shall have a maximum weight of 35 grams.

* 3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

* 3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

* 3.6.6 Depth. The module shall have no additional depth requirements.

* 3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	Output voltage (low-loaded)	III	S	0.01	0 8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B1	Input breakdown current (enable pin 37)	IV	S	0	2 0	0	2.0	0	2 0	0	2 0	0	2.0	mA
B2-B35	Input breakdown current	IV	S	0	1 0	0	1.0	0	1.0	0	1.0	0	1.0	mA
C1-C35	Input clamp diode voltage	V	S	-0 4	-1 5	-0.4	-1 5	-0.4	-1 5	-0.4	-1.5	-0.4	-1.5	V
D1-D48	Output voltage, high level	VI	X	2.4	4 5	2 4	4 5	2.4	4 5	2 4	4.5	2.4	4.5	V
E1-E72	Output voltage, low level													
	Initial	VII	X	0.00	0.40	0 00	0 40	0.00	0.40	0.00	0.40	0 00	0.40	V
	End-of-life	VII	X	0.00	0 40	0.00	0 40	0 00	0.40	0 00	0.42	0.00	0.42	V
F1	Input current, high level (enable pin 37)	VIII	X	0	80	0	80	0	80	0	80	0	80	µA
F2-F35	Input current, high level	VIII	X	0	40	0	40	0	40	0	40	0	40	µA
G1	Input current, low level (enable pin 37)	IX	X	0.20	3.46	0 20	3 46	0.20	3.46	0 20	3.46	0.20	3.46	mA
G2-G35	Input current, low level	IX	X	0.10	1.73	0.10	1.73	0 10	1.73	0.10	1.73	0.10	1.73	mA
H1-H3	Output short circuit current	X	S	20	100	20	115	20	115	20	115	20	115	mA
J1	Power supply current, low level	XI	X	48	132	48	132	48	132	48	132	48	132	mA
J2	Power supply current, high level	XI	X	48	132	48	132	48	132	48	132	48	132	mA
K	Isolation													
	Initial	XII	S	10	INF	10	INF	10	INF	10	INF	10	INF	MOhm
	End-of-life	XII	S	1.0	INF	1.0	INF	1.0	INF	1.0	INF	1.0	INF	MOhm
L1	Input capacitance (enable pin 37)	XIII	Q	4	30	4	30	4	30	4	30	4	30	pF
L2-L35	Input capacitance	XIII	Q	2	15	2	15	2	15	2	15	2	15	pF

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M3	Transition time (tTLH)													
	Initial	XIV	S	5	45	5	50	5	50	5	50	5	50	nS
	End-of-life	XIV	S	5	50	5	55	5	55	5	55	5	55	nS
N1-N3	Transition time (tTHL)													
	Initial	XV	S	5	25	5	30	5	30	5	30	5	30	nS
	End-of-life	XV	S	5	30	5	35	5	35	5	35	5	35	nS
P1-P24	Propagation delay time (tPLH) (a through h inputs)													
	Initial	XVI	S	5	40	5	45	5	45	5	45	5	45	nS
	End-of-life	XVI	S	5	40	5	45	5	45	5	45	5	45	nS
P25,P33, P41	Propagation delay time (tPLH) (enable inputs)	XVI	S	7	62	7	67	7	67	7	67	7	67	nS
P26-P32, P34-P40, P42-P48	Propagation delay time (tPLH) (enable inputs)	XVI	Q	7	62	7	67	7	67	7	67	7	67	nS
P51,P59, P71,P75, P83,P95, P99,P107, P119	Propagation delay time (tPLH) (A, B and C control inputs)	XVI	S	7	62	7	67	7	67	7	67	7	67	nS
P49,P50, P52-P58, P60-P70, P72-P74, P76-P82, P84-P94, P96-P98, P100-P106, P108-P118, P120	Propagation delay time (tPLH) (A, B and C control inputs)	XVI	Q	7	62	7	67	7	67	7	67	7	67	nS
Q1-Q24	Propagation delay time (tPHL) (a through h inputs)	XVII	S	5	40	5	45	5	45	5	45	5	45	nS
Q25,Q33, Q41	Propagation delay time (tPHL) (enable inputs)	XVII	S	7	50	7	55	7	55	7	55	7	55	nS

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/ - 3/	25° ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0° ± 5°C		60° ± 5°C		-55° ± 5°C		85° ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q26-Q32, Q34-Q40, Q42-Q48	Propagation delay time (tPHL) (enable inputs)	XVII	Q	7	50	7	55	7	55	7	55	7	55	nS
Q51,Q59, Q71,Q75, Q83,Q95, Q99,Q107, Q119	Propagation delay time (tPHL) (A, B and C control inputs)	XVII	S	7	50	7	55	7	55	7	55	7	55	nS
Q49,Q50, Q52-Q58, Q60-Q70, Q72-Q74, Q76-Q82, Q84-Q94, Q96-Q98, Q100-Q106, Q108-Q118, Q120	Propagation delay time (tPHL) (A, B and C control inputs)	XVII	Q	7	50	7	55	7	55	7	55	7	55	nS
R	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
S1-S3	Output crosstalk, high level	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
T1-T3	Output crosstalk, low level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply transient rise time	XXI	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µS

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25° C

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:22Z
Check the source to verify that this is the current version before use.

* 3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/5-1 for class I and M28787/5-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

* 3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

* 3.8 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

* 3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

* 3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 VDC, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 VDC ± 1 percent.

* 3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

* 4.4 Test procedures. The module shall be tested as specified in table I.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:22Z
Check the source to verify that this is the current version before use.

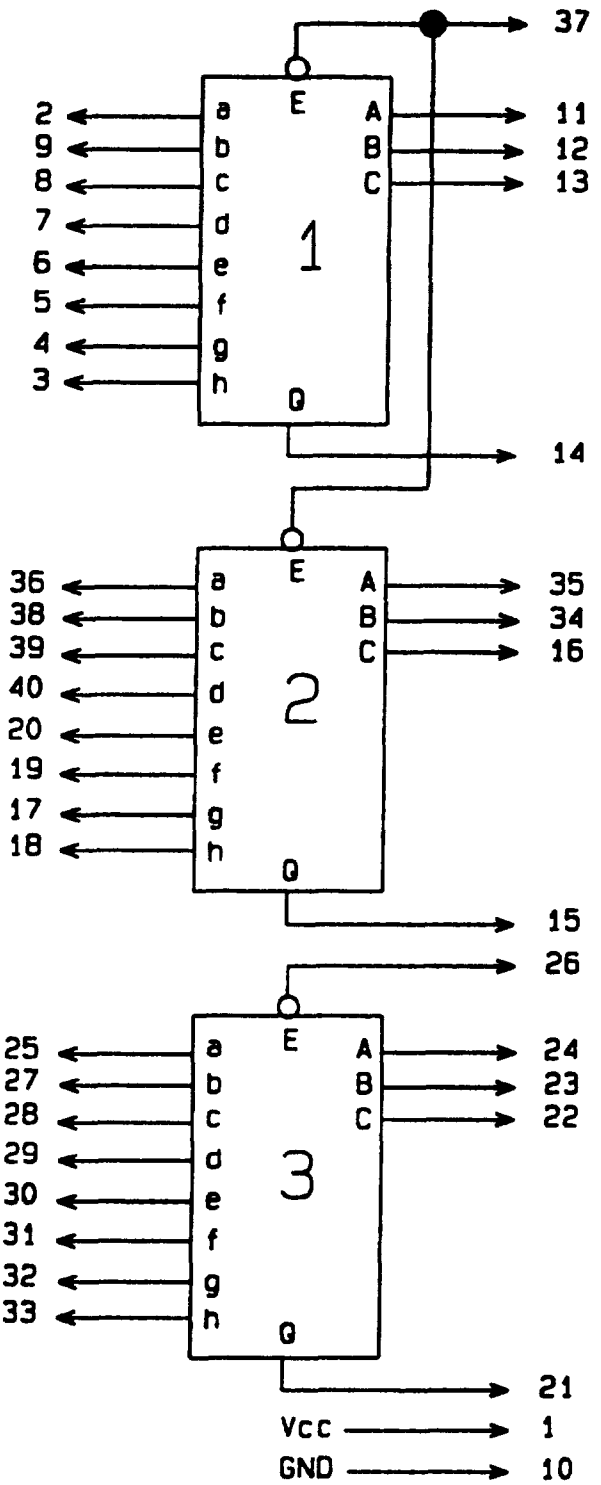


FIGURE 1. BDL module function diagram, triple 8-input digital multiplexer.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the current version before use.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through U shall be performed in any order with the following exceptions. Test conditions A, B, C, H and K are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, H or K are part of a sequence of tests, the order of testing shall be as follows: A, B, C, H and K where applicable, followed by the remaining tests in the sequence.

* 4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through G, J, M, N, P1-25, P33, P41, P51, P59, P71, P75, P83, P95, P99, P107, P119, Q1-Q25, Q33, Q41, Q51, Q59, Q71, Q75, Q83, Q95, Q99, Q107, Q109 and U shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

* 4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25 ± 5°C ambient.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:22Z
Check the source to verify that this is the currentversion before use.

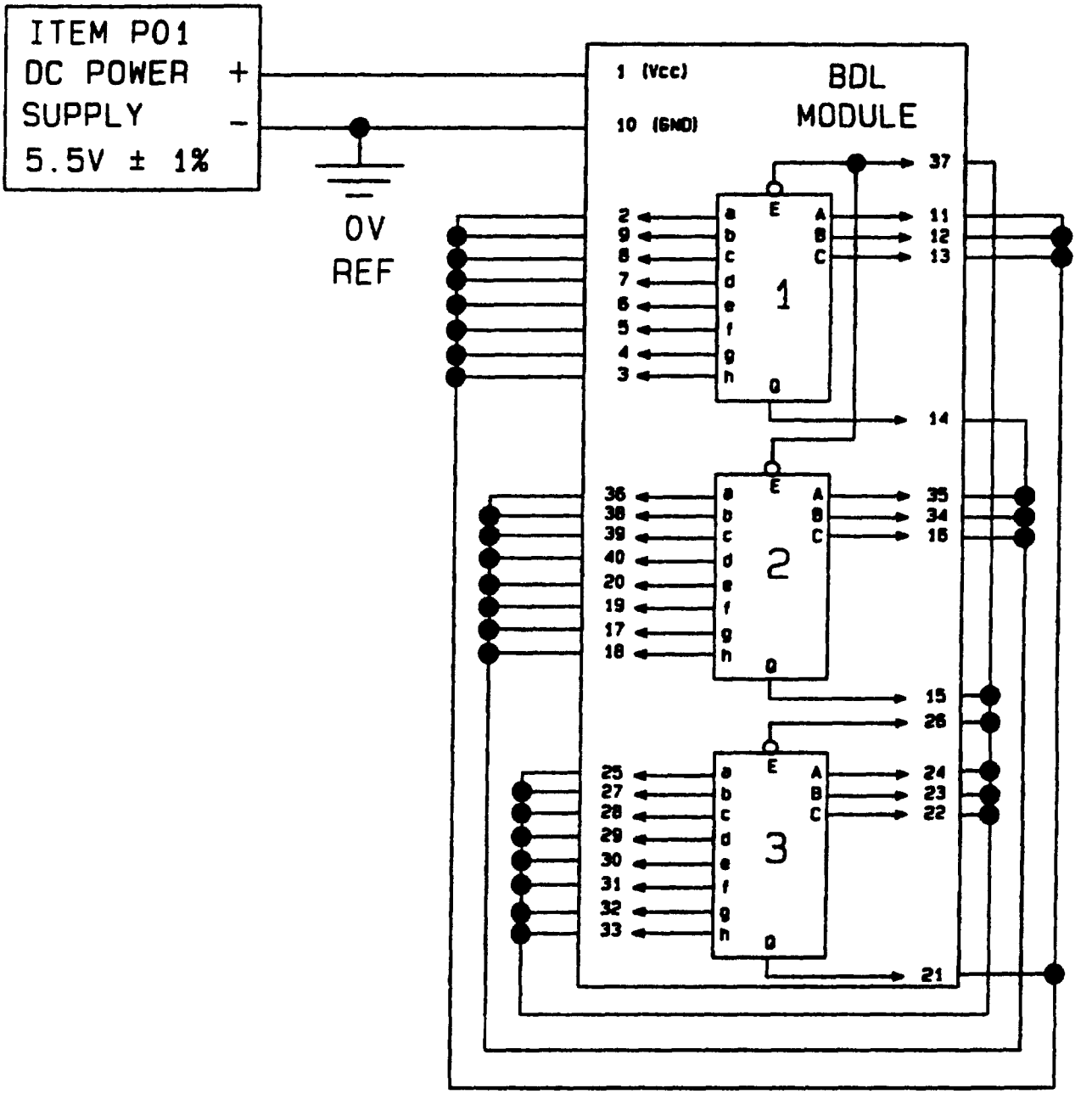
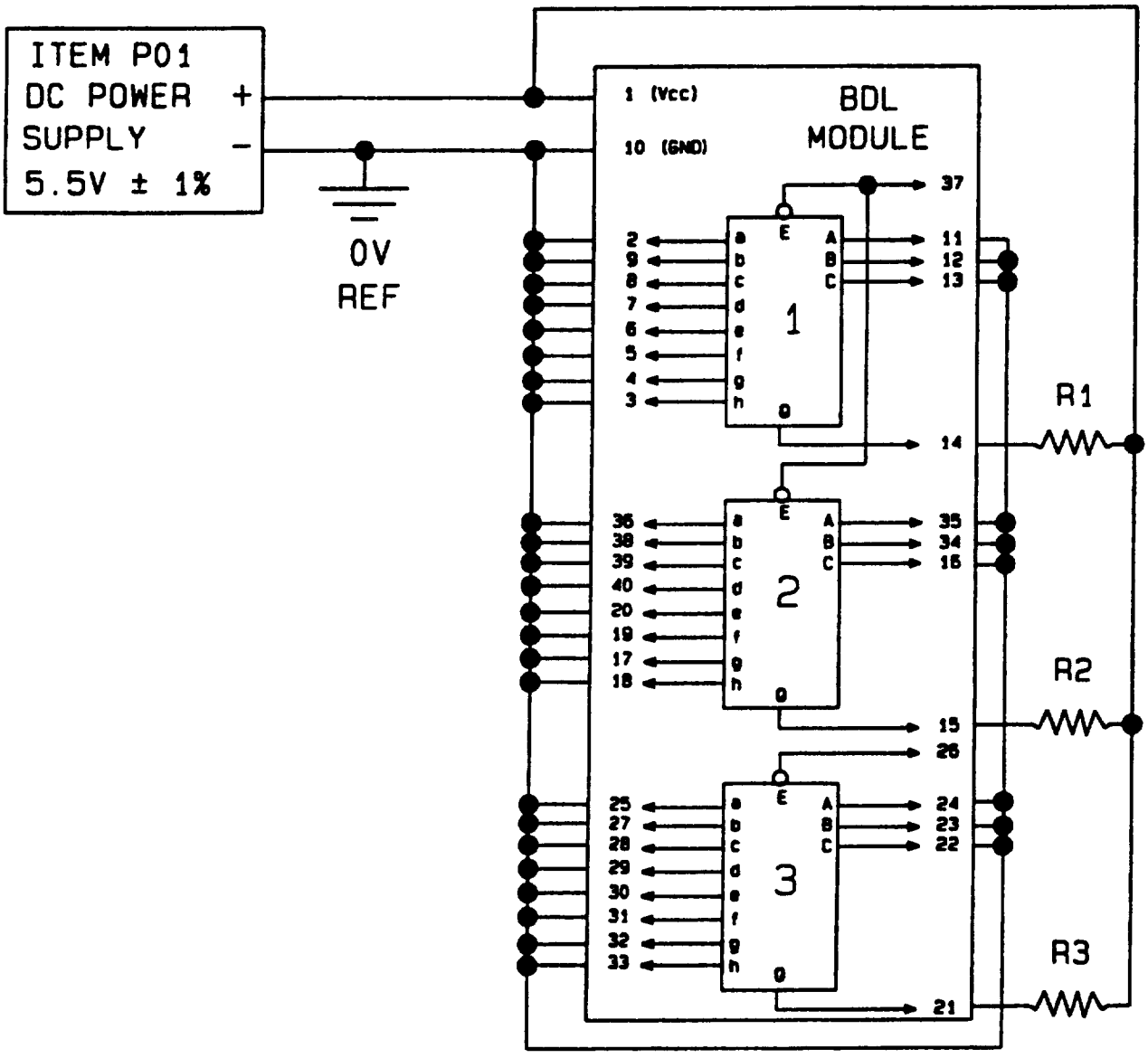


FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the currentversion before use.



NOTE: R1-R3 = 316 Ohms ± 1 percent, 1/4 W.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the current version before use.

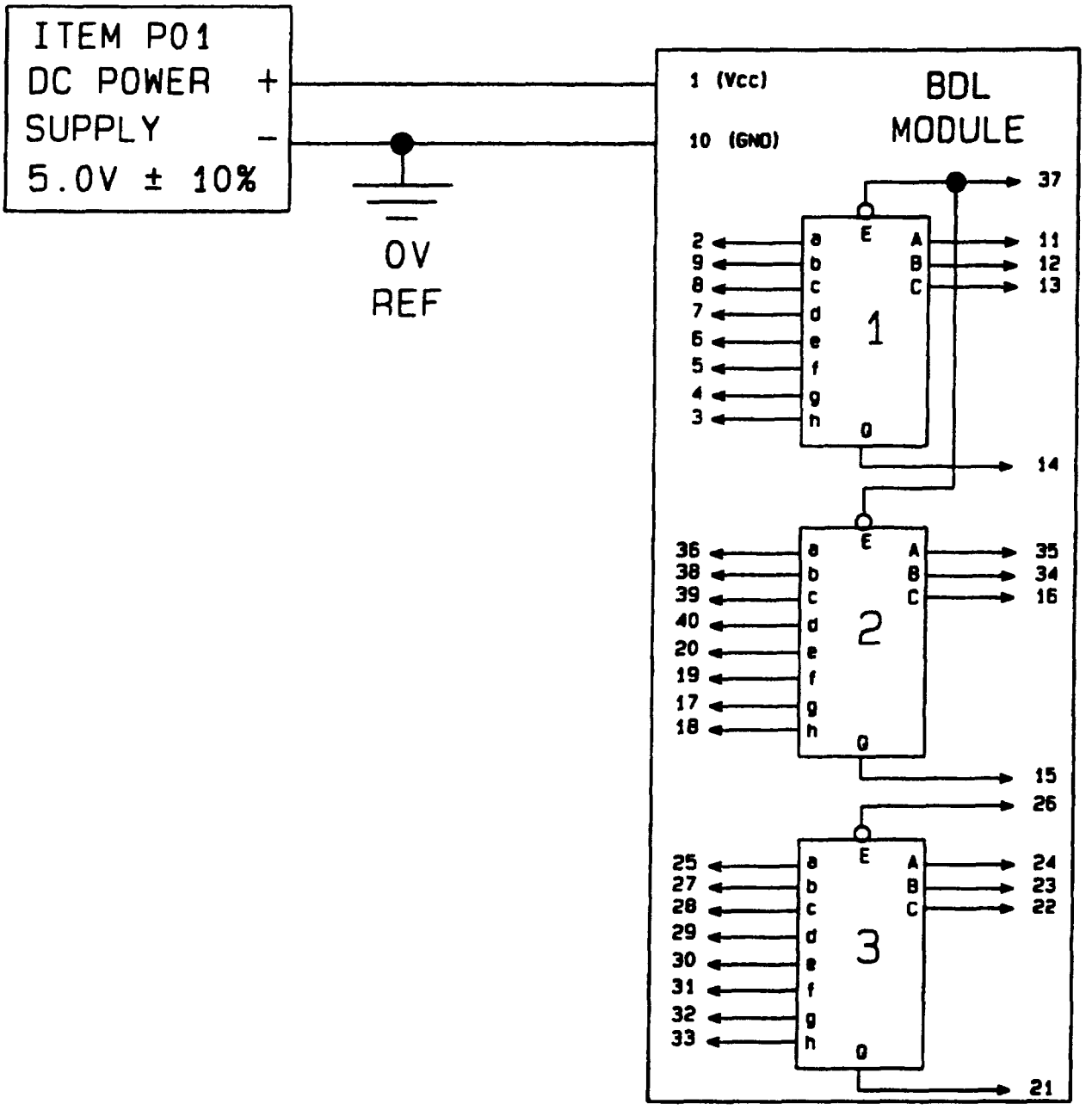


FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through G, J, M, N, P1-P25, P33, P41, P51, P59, P71, P75, P83, P95, P99, P107, P119, Q1-Q25, Q33, Q41, Q51, Q59, Q71, Q75, Q83, Q95, Q99, Q107, Q119 and U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests D through G, J, M, N, P1-P25, P33, P41, P51, P59, P71, P75, P83, P95, P99, P107, P119, Q1-Q25, Q33, Q41, Q51, Q59, Q71, Q75, Q83, Q95, Q99, Q107, Q119 and U shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

* 4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

* 6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/5 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658859. Part number M28787/5-1 supersedes part numbers 2658859 and M28787/5.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

Supply voltage	5 Vdc ± 10 percent
Maximum power dissipation	726 mW (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, Vcc maximum
Input current, a through h and A, B, C	
Low level (Vin = 0V)	-1.73 mA maximum
(Vin = 0.4 V)	-1.6 mA maximum
High level (Vin = 2.4 V)	40 µA maximum
(Vin = 5.5 V)	1.0 mA maximum
Input current (enable pin 37)	
Low level (VIL = 0 V)	-3.46 mA maximum
(VIL = 0.4 V)	-3.2 mA maximum
High level (VIH = 2.4 V)	80 µA maximum
(VIH = 5.5 V)	2.0 mA maximum
Input current (enable pin 26)	
Low level (VIL = 0 V)	-1.73 mA maximum
(VIL = 0.4 V)	-1.6 mA maximum
High level (VIH = 2.4 V)	40 µA maximum
(VIH = 5.5 V)	1.0 mA maximum
Output voltage	
Low level (IOL = 16 mA)	0.42 V maximum
High level (IOH = -800 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (VOL = 0.4 V)	16 mA
High level (VOH = 2.4 V)	-800 µA

* 6.5 Module function. The multiplexer module, designated as key code BDL, is a Standard Electronic Module (SEM) with 0.100 inch (2.54 mm) grid center contacts. This module contains three 8-input multiplexers. Each multiplexer has eight input data lines (a through h), three input data control lines (A, B and C) and an enable input line (E). The data line used is selected by the state of the data control inputs. This data is transferred to the output (Q) providing the enable input is low. When it is high, the output is low regardless of all other inputs. The enable input on multiplexers 1 and 2 is tied to the same module pin (see figure 1). Logic is positive with each multiplexer output as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

$$Q = \overline{E} \cdot (ABCa + \overline{A}BCb + A\overline{B}Cc + \overline{AB}Cd + AB\overline{C}e + \overline{A}\,\overline{B}Cf + A\overline{BC}g + \overline{ABC}h)$$

* 6.5.1 Input to output relationship. Each 8-input multiplexer has the input to output relationship in table II for positive logic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE II. Input to output relationship.

Inputs			Output
Control	Enable	Data input lines	
C B A	E	h g f e d c b a	Q
X X X	H	X X X X X X X X	L
L L L	L	L X X X X X X X	L
L L L	L	H X X X X X X X	H
L L H	L	X L X X X X X X	L
L L H	L	X H X X X X X X	H
L H L	L	X X L X X X X X	L
L H L	L	X X H X X X X X	H
L H H	L	X X X L X X X X	L
L H H	L	X X X H X X X X	H
H L L	L	X X X X L X X X	L
H L L	L	X X X X H X X X	H
H L H	L	X X X X X L X X	L
H L H	L	X X X X X H X X	H
H H L	L	X X X X X X L X	L
H H L	L	X X X X X X H X	H
H H H	L	X X X X X X X L	L
H H H	L	X X X X X X X H	H

$Q = \overline{E}\,(ABCa + \overline{A}BCb + A\overline{B}Cc + \overline{AB}Cd + AB\overline{C}e + \overline{A}B\overline{C}f + A\overline{BC}g + \overline{ABC}h)$

X = Level does not affect output.

L = Low voltage level.

H = High voltage level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* 6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*Custodians:
Army - ER
Navy - NW
Air Force - 85

Review activities:
Army - AT
Navy - MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

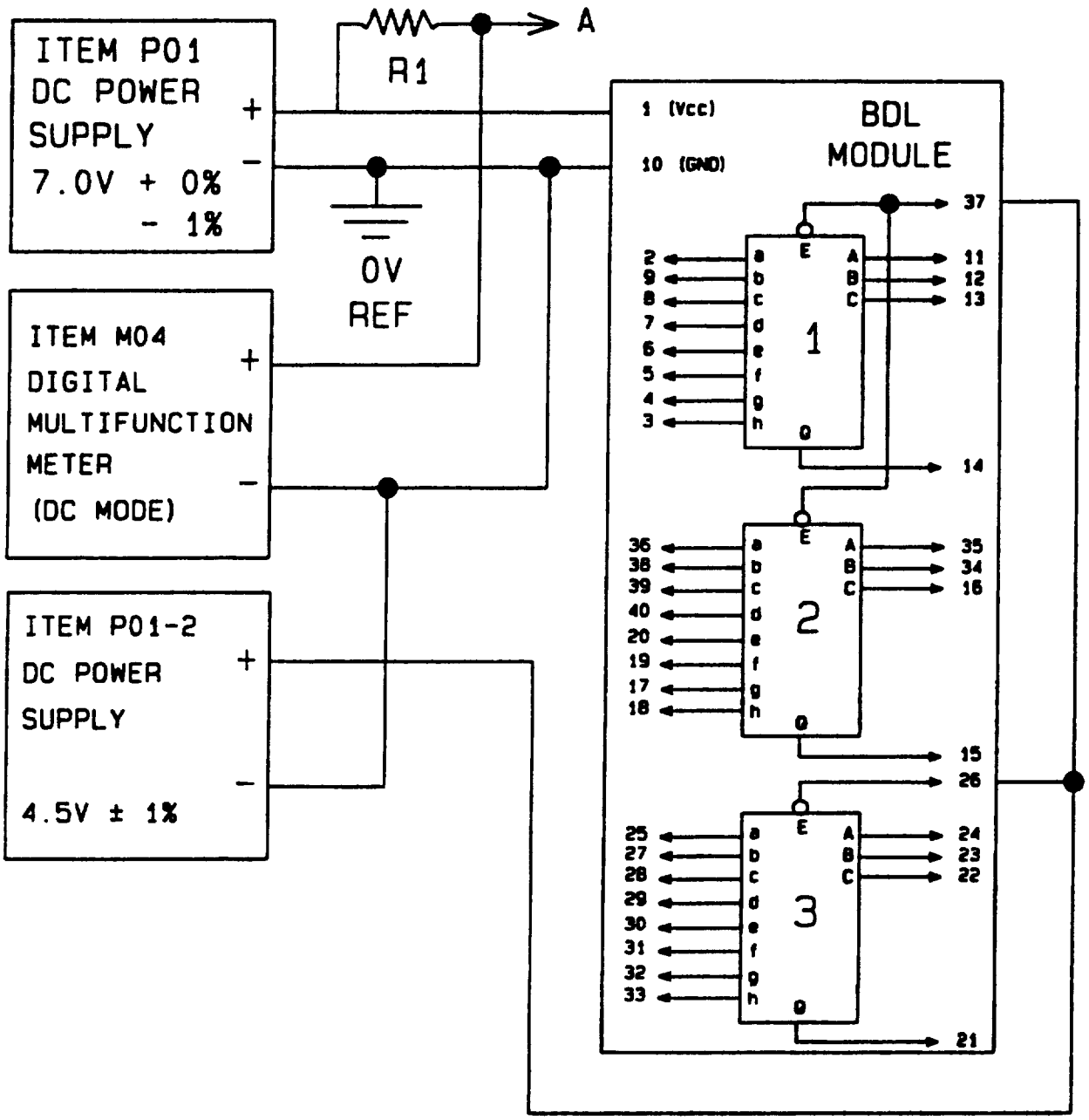
(Project 5963-0083)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE III. Output voltage (low loaded) test.

Test cond.	Parameter	Connect	Initial limits 25° ± 5° C Ambient		Unit	Test Figure
		A to pin	Min	Max		
A1	Q1	14	0.01	0.8	V	5
A2	Q2	15				
A3	Q3	21	0.01	0.8	V	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



NOTES:

1. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.

2. R1 = 316 Ohms ± 1 percent, 1/4 W.

* FIGURE 5. Output voltage (low loaded) test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE IV. Input breakdown current test.

Test cond.	Parameter	Connect			Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
B1	E1, E2 (enable)	37	Open	11, 12, 13, 16, 34, 35	0	2.0	mA	6
B2	A1 (A control)	11		12, 13, 37		1.0		
B3	B1 (B control)	12		11, 13, 37				
B4	C1 (C control)	13	Open	11, 12, 37				
B5	a1 (a input)	2	11, 12, 13	37				
B6	b1 (b input)	9	12, 13	11, 37				
B7	c1 (c input)	8	11, 13	12, 37				
B8	d1 (d input)	7	13	11, 12, 37				
B9	e1 (e input)	6	11, 12	13, 37				
B10	f1 (f input)	5	12	11, 13, 37	0	1.0	mA	6

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE IV. Input breakdown current test - Continued.

Test cond.	Parameter	Connect			Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
B11	g1 (g input)	4	11	12, 13, 37	0	1.0	mA	6
B12	h1 (h input)	3	Open	11, 12, 13, 37				
B13	A2 (A control)	35		16, 34, 37				
B14	B2 (B control)	34		16, 35, 37				
B15	C2 (C control)	16	Open	34, 35, 37				
B16	a2 (a input)	36	16, 34, 35	37				
B17	b2 (b input)	38	16, 34	35, 37				
B18	c2 (c input)	39	16, 35	34, 37				
B19	d2 (d input)	40	16	34, 35, 37				
B20	e2 (e input)	20	34, 35	16, 37				
B21	f2 (f input)	19	34	16, 35, 37	0	1.0	mA	6

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE IV. Input breakdown current test - Continued.

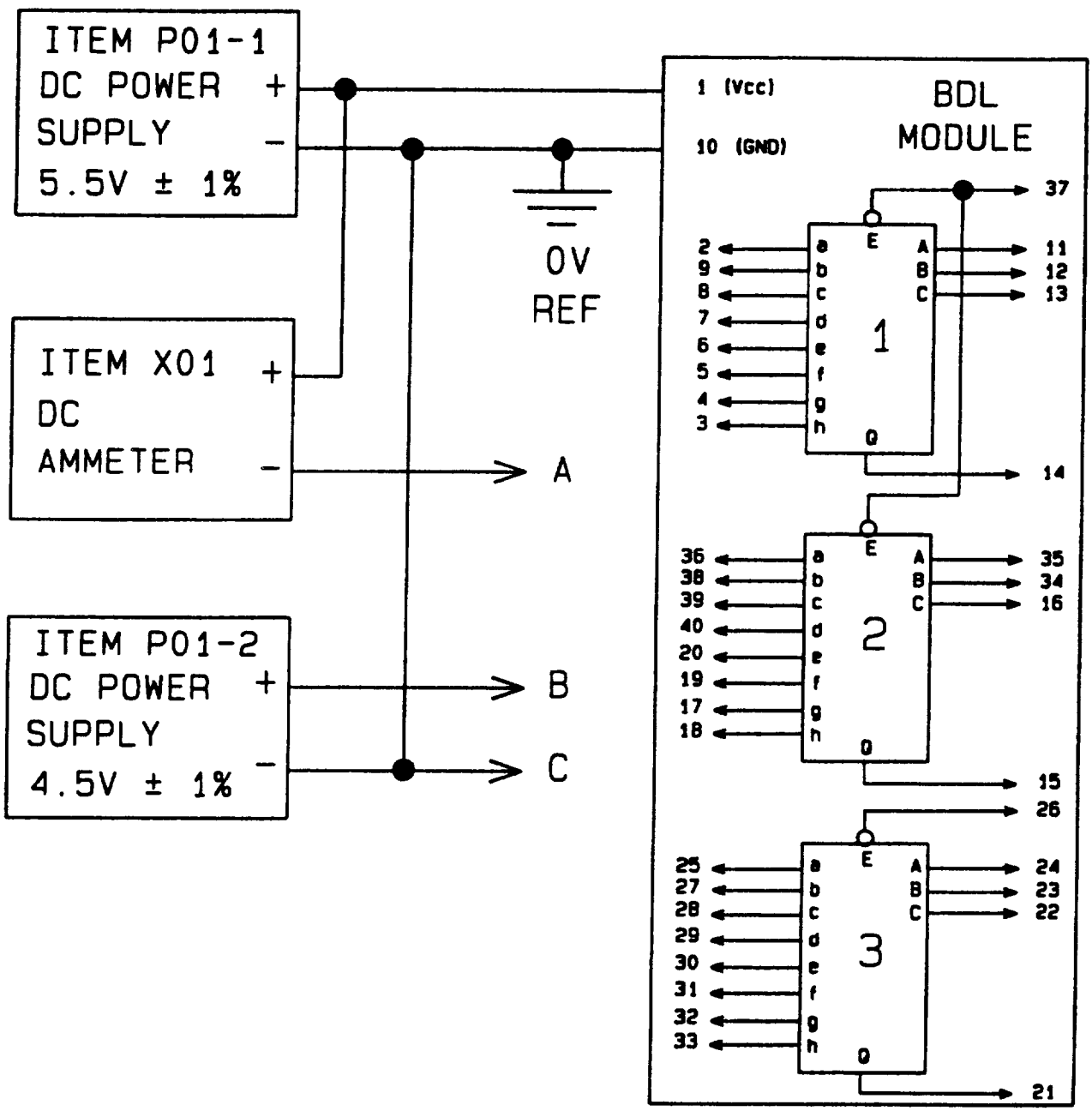
Test cond.	Parameter	Connect			Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
B22	g2 (g input)	17	35	16, 34, 37	0	1.0	mA	6
B23	h2 (h input)	18	Open	16, 34, 35, 37				
B24	E3 (enable)	26		22, 23, 24				
B25	A3 (A control)	24		22, 23, 26				
B26	B3 (B control)	23		22, 24, 26				
B27	C3 (C control)	22	Open	23, 24 26				
B28	a3 (a input)	25	22, 23, 24	26				
B29	b3 (b input)	27	22, 23	24,26				
B30	c3 (c input)	28	22, 24	23, 26				
B31	d3 (d input)	29	22	23, 24, 26				
B32	e3 (e input)	30	23, 24	22, 26	0	1.0	mA	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE IV. Input breakdown current test - Continued.

Test cond.	Parameter	Connect			Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Min	Max		
B33	f3 (f input)	31	23	22, 24, 26	0	1.0	mA	6
B34	g3 (g input)	32	24	22, 23, 26				
B35	h3 (h input)	33	Open	22, 23 24, 26	0	1.0	mA	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22
Check the source to verify that this is the current version before use.



CAUTION: Do not isolate chassis of ammeter X01 from earth ground.

FIGURE 6. Input breakdown current test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test.

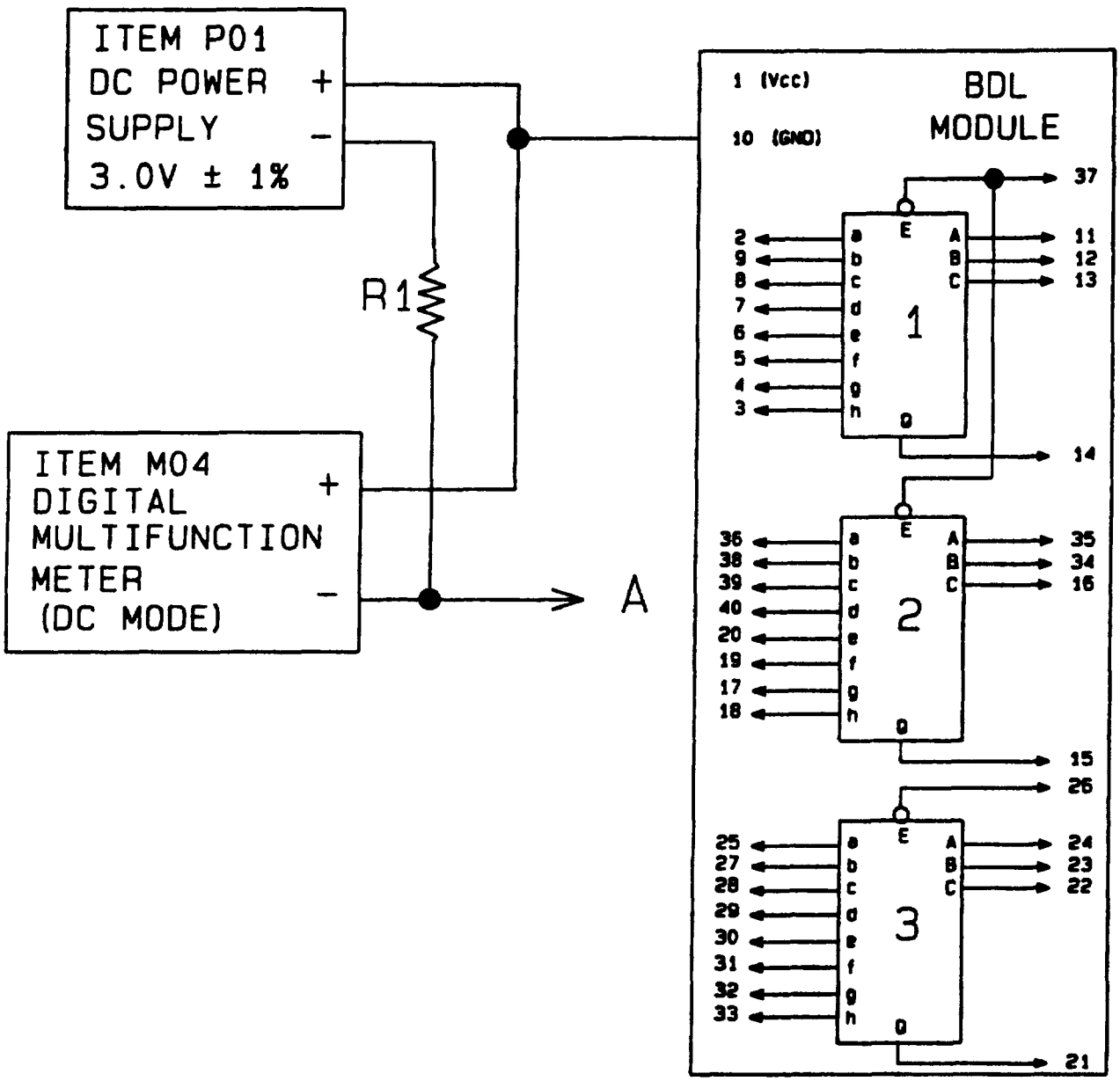
Test cond.	Parameter	Connect A to pin	Initial limits 25° ± 5° C Ambient Min	Initial limits 25° ± 5° C Ambient Max	Unit	Test fig.
C1	E1 and E2 (enable pin 37)	37	-0.4	-1.5	V	7
C2	A1 (A control)	11				
C3	B1 (B control)	12				
C4	C1 (C control)	13				
C5	a1 (a input)	2				
C6	b1 (b input)	9				
C7	c1 (c input)	8				
C8	d1 (d input)	7				
C9	e1 (e input)	6				
C10	f1 (f input)	5				
C11	g1 (g input)	4				
C12	h1 (h input)	3				
C13	A2 (A control)	35				
C14	B2 (B control)	34				
C15	C2 (C control)	16				
C16	a2 (a input)	36				
C17	b2 (b input)	38				
C18	c2 (c input)	39				
C19	d2 (d input)	40				
C20	e2 (e input)	20	-0.4	-1.5	V	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test - Continued.

Test cond.	Parameter	Connect	Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	Min	Max		
C21	f2 (f input)	19	-0.4	-1.5	V	7
C22	g2 (g input)	17				
C23	h2 (h input)	18				
C24	E3 (enable)	26				
C25	A3 (A control)	24				
C26	B3 (B control)	23				
C27	C3 (C control)	22				
C28	a3 (a input)	25				
C29	b3 (b input)	27				
C30	c3 (c input)	28				
C31	d3 (d input)	29				
C32	e3 (e input)	30				
C33	f3 (f input)	31				
C34	g3 (g input)	32				
C35	h3 (h input)	33	-0.4	-1.5	V	7

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 124 Ohms ± 1 percent, 1/4 W.

* FIGURE 7. Input clamp diode voltage test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 20261 0-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VI Output voltage, high level, test

Test cond	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
D1	Q1 (a input), A, B and C threshold "H"; E threshold "L"	11, 12, 13	37	14	2	3, 4, 5, 6, 7, 8, 9	2.4	4.5	V	8
D2	Q1 (b input), B and C threshold "H"; A and E threshold "L"	12, 13	11, 37		9	2, 3, 4, 5, 6, 7, 8				
D3	Q1 (c input), A and C threshold "H"; B and E threshold "L"	11, 13	12, 37		8	2, 3, 4, 5, 6, 7, 9				
D4	Q1 (d input), C threshold "H"; A, B and E threshold "L"	13	11, 12, 37		7	2, 3, 4, 5, 6, 8, 9				
D5	Q1 (e input), A and B threshold "H"; C and E threshold "L"	11, 12	13, 37		6	2, 3, 4, 5, 7, 8, 9				
D6	Q1 (f input), B threshold "H"; A, C and E threshold "L"	12	11, 13, 37		5	2, 3, 4, 6, 7, 8, 9				
D7	Q1 (g input), A threshold "H"; B, C and E threshold "L"	11	12, 13, 37		4	2, 3, 5, 6, 7, 8, 9				
D8	Q1 (h input), A, B, C and E threshold "L"	Open	11, 12, 13, 37	14	3	2, 4, 5, 6, 7, 8, 9				
D9	Q2 (a input), A, B and C threshold "H"; E threshold "L"	16, 34, 35	37	15	36	17, 18, 19, 20, 38, 39, 40				
D10	Q2 (b input), B and C threshold "H"; A and E threshold "L"	16, 34	35, 37	15	38	17, 18, 19, 20, 36, 39, 40	2.4	4.5	V	8

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage, high level, test - Continued.

Test cond.	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
D11	Q2 (c input), A and C threshold "H"; B and E threshold "L"	16, 35	34, 37	15	39	17, 18, 19, 20, 36, 38, 40	2.4	4.5	V	8
D12	Q2 (d input), C threshold "H"; A, B and E threshold "L"	16	34, 35, 37		40	17, 18, 19, 20, 36, 38, 39				
D13	Q2 (e input), A and B threshold "H"; C and E threshold "L"	34, 35	16, 37		20	17, 18, 19, 36, 38, 39, 40				
D14	Q2 (f input), B threshold "H"; A, C and E threshold "L"	34	16, 35, 37		19	17, 18, 20, 36, 38, 39, 40				
D15	Q2 (g input), A threshold "H"; B, C and E threshold "L"	35	16, 34, 37		17	18, 19, 20, 36, 38, 39, 40				
D16	Q2 (h input), A, B, C and E threshold "L"	Open	16, 34, 35, 37	15	18	17, 19, 20, 36, 38, 39, 40				
D17	Q3 (a input), A, B and C threshold "H"; E threshold "L"	22, 23, 24	26	21	25	27, 28, 29, 30, 31, 32, 33				
D18	Q3 (b input), B and C threshold "H"; A and E threshold "L"	22, 23	24, 26		27	25, 28, 29, 30, 31, 32, 33				
D19	Q3 (c input), A and C threshold "H"; B and E threshold "L"	22, 24	23, 26		28	25, 27, 29, 30, 31, 32, 33				
D20	Q3 (d input), C threshold "H"; A, B and E threshold "L"	22	23, 24, 26	21	29	25, 27, 28, 30, 31, 32, 33	2.4	4.5	V	8

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage, high level, test - Continued.

Test cond.	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
D21	Q3 (e input), A and B threshold "H"; C and E threshold "L"	23, 24	22, 26	21	30	25, 27, 28, 29, 31, 32, 33	2.4	4.5	V	8
D22	Q3 (f input), B threshold "H"; A, C and E threshold "L"	23	22, 24, 26		31	25, 27, 28, 29, 30, 32, 33				
D23	Q3 (g input), A threshold "H"; B, C and E threshold "L"	24	22, 23, 26		32	25, 27, 28, 29, 30, 31, 33				
D24	Q3 (h input), A, B, C and E threshold "L"	Open	22, 23, 24, 26	21	33	25, 27, 28, 29, 30, 31, 32				
D25	Q1, a threshold "H"	2	Open	14	11, 12, 13	37				
D26	Q1, b threshold "H"	9			12, 13	11, 37				
D27	Q1, c threshold "H"	8			11, 13	12, 37				
D28	Q1, d threshold "H"	7			13	11, 12, 37				
D29	Q1, e threshold "H"	6			11, 12	13, 37				
D30	Q1, f threshold "H"	5			12	11, 13, 37				
D31	Q1, g threshold "H"	4			11	12, 13, 37				
D32	Q1, h threshold "H"	3		14	Open	11, 12, 13, 37				
D33	Q2, a threshold "H"	36		15	16, 34, 35	37				
D34	Q2, b threshold "H"	38			16, 34	35, 37				
D35	Q2, c threshold "H"	39			16, 35	34, 37				
D36	Q2, d threshold "H"	40			16	34, 35, 37				
D37	Q2, e threshold "H"	20	Open	15	34, 35	16, 37	2.4	4.5	V	8

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage, high level, test - Continued.

Test cond.	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
D38	Q2, f threshold "H"	19	Open	15	34	16, 35, 37	2.4	4.5	V	8
D39	Q2, g threshold "H"	17			35	16, 34, 37				
D40	Q2, h threshold "H"	18		15	Open	16, 34, 35 37				
Q41	Q3, a threshold "H"	25		21	22, 23, 24	26				
D42	Q3, b threshold "H"	27			22, 23	24, 26				
D43	Q3, c threshold "H"	28			22, 24	23, 26				
D44	Q3, d threshold "H"	29			22	23, 24, 26				
D45	Q3, e threshold "H"	30			23, 24	22, 26				
D46	Q3, f threshold "H"	31			23	22, 24, 26				
D47	Q3, g threshold "H"	32			24	22, 23, 26				
D48	Q3, h threshold "H"	33	Open	21	Open	22, 23, 24, 26	2.4	4.5	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VII Output voltage, low level, test.

Test cond	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E1	Q1 (a input), A, B and C threshold "H"; E threshold "L"	11, 12, 13	37	14	3, 4, 5, 6, 7, 8, 9	2	0.00	0.40	V	8
E2	Q1 (b input), B and C threshold "H"; A and E threshold "L"	12, 13	11, 37		2, 3, 4, 5, 6, 7, 8	9				
E3	Q1 (c input), A and C threshold "H"; B and E threshold "L"	11, 13	12, 37		2, 3, 4, 5, 6, 7, 9	8				
E4	Q1 (d input), C threshold "H"; A, B and E threshold "L"	13	11, 12, 37		2, 3, 4, 5, 6, 8, 9	7				
E5	Q1 (e input), A and B threshold "H"; C and E threshold "L"	11, 12	13, 37		2, 3, 4, 5, 7, 8, 9	6				
E6	Q1 (f input), B threshold "H"; A, C and E threshold "L"	12	11, 13, 37		2, 3, 4, 6, 7, 8, 9	5				
E7	Q1 (g input), A threshold "H"; B, C and E threshold "L"	11	12, 13, 37		2, 3, 5, 6, 7, 8, 9	4				
E8	Q1 (h input), A, B, C and E threshold "L"	Open	11, 12, 13, 37	14	2, 4, 5, 6, 7, 8, 9	3				
E9	Q2 (a input), A, B and C threshold "H"; E threshold "L"	16, 34, 35	37	15	17, 18, 19, 20, 38, 39, 40	36				
E10	Q2 (b input), B and C threshold "H"; A and E threshold "L"	16, 34	35, 37	15	17, 18, 19, 20, 36, 39, 40	38	0.00	0.40	V	8

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VII. Output voltage, low level, test - Continued.

Test cond	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E11	Q2 (c input), A and C threshold "H"; B and E threshold "L"	16, 35	34, 37	15	17, 18, 19, 20, 36, 38, 40	39	0.00	0.40	V	8
E12	Q2 (d input), C threshold "H"; A, B and E threshold "L"	16	34, 35, 37		17, 18, 19, 20, 36, 38, 39	40				
E13	Q2 (e input), A and B threshold "H"; C and E threshold "L"	34, 35	16, 37		17, 18, 19, 36, 38, 39, 40	20				
E14	Q2 (f input), B threshold "H"; A, C and E threshold "L"	34	16, 35, 37		17, 18, 20, 36, 38, 39, 40	19				
E15	Q2 (g input), A threshold "H"; B, C and E threshold "L"	35	16, 34, 37		18, 19, 20, 36, 38, 39, 40	17				
E16	Q2 (h input), A, B, C and E threshold "L"	Open	16, 34, 35, 37	15	17, 19, 20, 36, 38, 39, 40	18				
E17	Q3 (a input), A, B and C threshold "H"; E threshold "L"	22, 23, 24	26	21	27, 28, 29, 30, 31, 32, 33	25				
E18	Q3 (b input), B and C threshold "H"; A and E threshold "L"	22, 23	24, 26		25, 28, 29, 30, 31, 32, 33	27				
E19	Q3 (c input), A and C threshold "H"; B and E threshold "L"	22, 24	23, 26		25, 27, 29, 30, 31, 32, 33	28				
E20	Q3 (d input), C threshold "H"; A, B and E threshold "L"	22	23, 24, 26	21	25, 27, 28, 30, 31, 32, 33	29	0.00	0.40	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VII Output voltage, low level, test - Continued.

Test cond.	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E21	Q3 (e input), A and B threshold "H"; C and E threshold "L"	23, 24	22, 26	21	25, 27, 28, 29, 31, 32, 33	30	0.00	0.40	V	8
E22	Q3 (f input), B threshold "H"; A, C and E threshold "L"	23	22, 24, 26		25, 27, 28, 29, 30, 32, 33	31				
E23	Q3 (g input), A threshold "H"; B, C and E threshold "L"	24	22, 23, 26		25, 27, 28, 29, 30, 31, 33	32				
E24	Q3 (h input), A, B, C and E threshold "L"	Open	22, 23, 24, 26	21	25, 27, 28, 29, 30, 31, 32	33				
E25	Q1 (a input), E threshold "H"	37	Open	14	2, 11, 12, 13	Open				
E26	Q1 (b input), E threshold "H"				9, 12, 13	11				
E27	Q1 (c input), E threshold "H"				8, 11, 13	12				
E28	Q1 (d input), E threshold "H"				7, 13	11, 12				
E29	Q1 (e input), E threshold "H"				6, 11, 12	13				
E30	Q1 (f input), E threshold "H"				5, 12	11, 13				
E31	Q1 (g input), E threshold "H"				4, 11	12, 13				
E32	Q1 (h input), E threshold "H"			14	3	11, 12, 13				
E33	Q2 (a input), E threshold "H"	37	Open	15	16, 34, 35, 36	Open	0.00	0.40	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VII Output voltage, low level, test - Continued.

Test cond.	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E34	Q2 (b input), E threshold "H"	37	Open	15	18, 34, 38	35	0.00	0.40	V	8
E35	Q2 (c input), E threshold "H"				18, 35, 39	34				
E36	Q2 (d input), E threshold "H"				18, 40	34, 35				
E37	Q2 (e input), E threshold "H"				20, 34, 35	18				
E38	Q2 (f input), E threshold "H"				19, 34	18, 35				
E39	Q2 (g input), E threshold "H"				17, 35	18, 34				
E40	Q2 (h input), E threshold "H"	37		15	18	18, 34, 35				
E41	Q3 (a input), E threshold "H"	26		21	22, 23, 24, 25	Open				
E42	Q3 (b input), E threshold "H"				22, 23, 27	24				
E43	Q3 (c input), E threshold "H"				22, 24, 28	23				
E44	Q3 (d input), E threshold "H"				22, 29	23, 24				
E45	Q3 (e input), E threshold "H"				23, 24, 30	22				
E46	Q3 (f input), E threshold "H"				23, 31	22, 24				
E47	Q3 (g input), E threshold "H"	26	Open	21	24, 32	22, 23	0.00	0.40	V	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VII. Output voltage, low level, test - Continued.

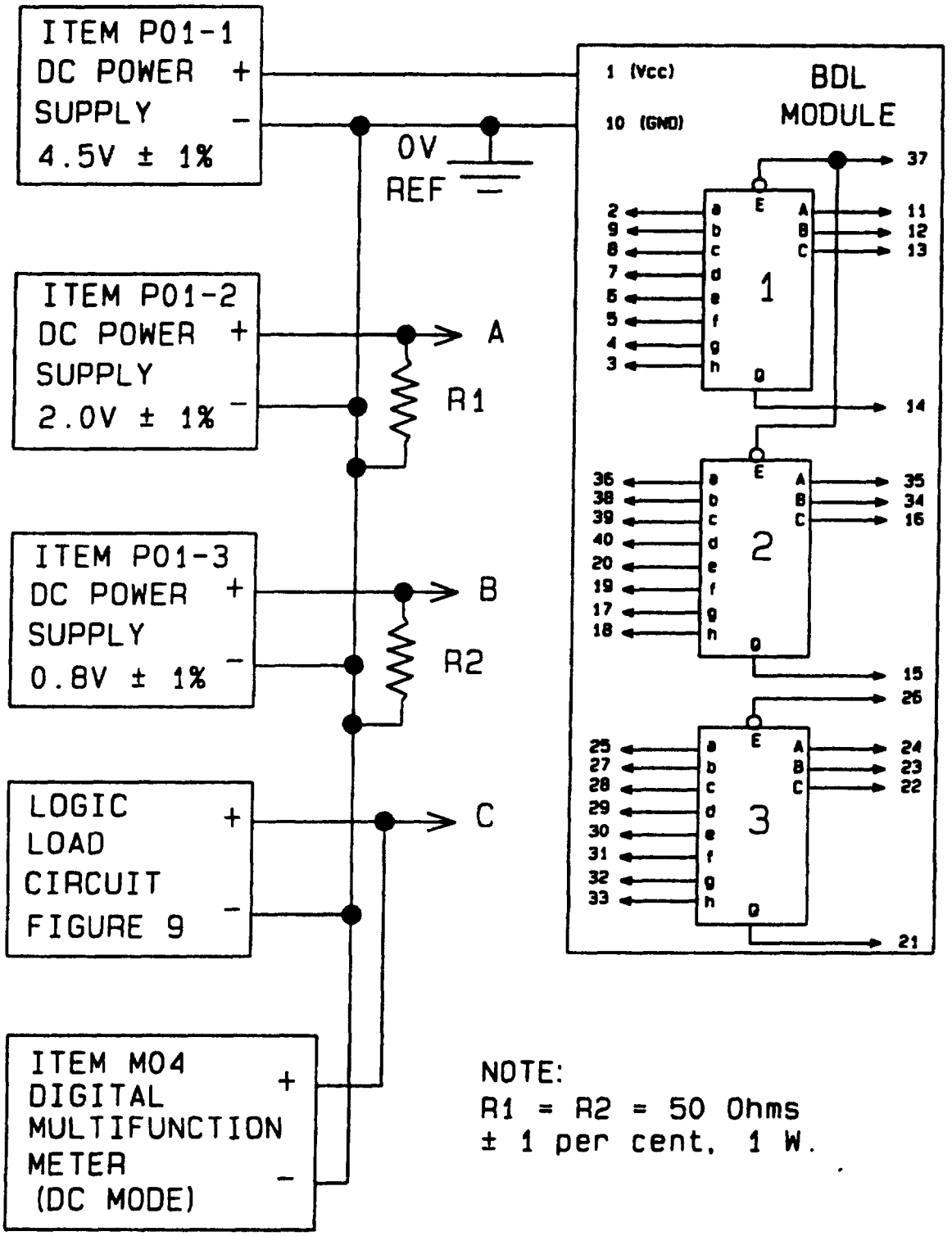
Test cond	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E48	Q3 (h input), E threshold "H"	26	Open	21	33	22, 23, 24	0.00	0.40	V	8
E49	Q1, a threshold "L"	Open	2	14	11, 12, 13	37				
E50	Q1, b threshold "L"		9		12, 13	11, 37				
E51	Q1, c threshold "L"		8		11, 13	12, 37				
E52	Q1, d threshold "L"		7		13	11, 12, 37				
E53	Q1, e threshold "L"		6		11, 12	13, 37				
E54	Q1, f threshold "L"		5		12	11, 13, 37				
E55	Q1, g threshold "L"		4		11	12, 13, 37				
E56	Q1, h threshold "L"		3	14	Open	11, 12, 13, 37				
E57	Q2, a threshold "L"		36	15	16, 34, 35	37				
E58	Q2, b threshold "L"		38		16, 34	35, 37				
E59	Q2, c threshold "L"		39		16, 35	34, 37				
E60	Q2, d threshold "L"		40		16	34, 35, 37				
E61	Q2, e threshold "L"		20		34, 35	16, 37				
E62	Q2, f threshold "L"		19		14	16, 35, 37				
E63	Q2, g threshold "L"		17		35	16, 34, 37				
E64	Q2, h threshold "L"		18	15	Open	16, 34, 35 37				
Q65	Q3, a threshold "L"		25	21	22, 23, 24	26				
E66	Q3, b threshold "L"		27		22, 23	24, 26				
E67	Q3, c threshold "L"	Open	28	21	22, 24	23, 26	0.00	0.40	V	8

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE VII. Output voltage, low level, test - Continued.

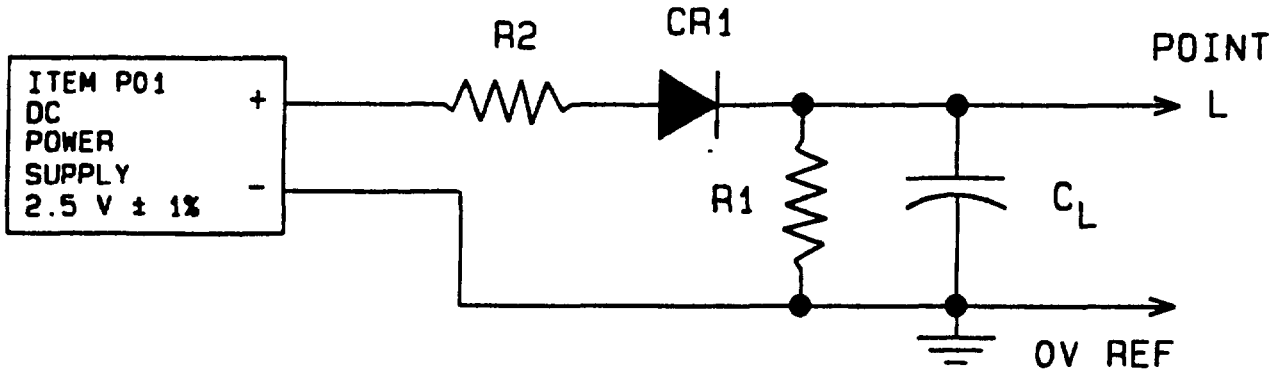
Test cond	Parameter	Connect					Initial limits 25° ± 5° C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
E68	Q3, d threshold "L"	Open	29	21	22	23, 24, 26	0.00	0.40	V	8
E69	Q3, e threshold "L"		30		23, 24	22, 26				
E70	Q3, f threshold "L"		31		23	22, 24, 26				
E71	Q3, g threshold "L"		32		24	22, 23, 26				
E72	Q3, h threshold "L"	Open	33	21	Open	22, 23, 24, 26	0.00	0.40	V	8

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.



* **FIGURE 8.** Output voltage (high Level and low level) test circuit for tables VI and VII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:22Z
Check the source to verify that this is the current version before use.



NOTES:

1. CR1 = 1N4150 or equivalent.
2. CL = 130 ± 5 pF (including probe and parasitic capacitance) for test conditions M, N, P and Q.
3. R1 = 3.01 kOhms ± 1 percent, 1/8 W.
4. R2 = Select value for a current flow of 16.0 mA ± 5 percent with point "L" held at 0.4 volt.
5. Capacitor CL is not required for test conditions E or F.
6. The maximum lead length from point "L" to the appropriate module pin is 3 inches (76.2 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV peak.
8. CL = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions S and T.

FIGURE 9. Logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

• TABLE VIII. Input current, high level, test.

Test cond	Parameter	Connect				Read meter	Initial limits 25° ± 5°C Ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin		Min	Max		
F1	E1 and E2 (enable pin 37), high level	37	Open	Open	11, 12, 13, 16, 34, 35	X01-1	0	80	µA	10
F2	A1 (A control), high level	11			12, 13, 37			40		
F3	B1 (B control), high level	12			11, 13, 37					
F4	C1 (C control), high level	13		Open	11, 12, 37					
F5	a1 (a input), high level	2		11, 12, 13	37					
F6	b1 (b input), high level	9		12, 13	11, 37					
F7	c1 (c input), high level	8		11, 13	12, 37					
F8	d1 (d input), high level	7		13	11, 12, 37					
F9	e1 (e input), high level	6		11, 12	13, 37					
F10	f1 (f input), high level	5		12	11, 13, 37					
F11	g1 (g input), high level	4		11	12, 13, 37					
F12	h1 (h input), high level	3		Open	11, 12, 13, 37					
F13	A2 (A control), high level	35			16, 34, 37					
F14	B2 (B control), high level	34	Open	Open	16, 35, 37	X01-1	0	40	µA	10

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE VIII. Input current, high level, test - Continued.

Test cond.	Parameter	Connect				Read meter	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin		Min	Max		
F15	C2 (C control), high level	16	Open	Open	34, 35, 37	X01-1	0	40	µA	10
F16	a2 (a input), high level	36		16, 34, 35	37					
F17	b2 (b input), high level	38		16, 34	35, 37					
F18	c2 (c input), high level	39		16, 35	34, 37					
F19	d2 (d input), high level	40		16	34, 35, 37					
F20	e2 (e input), high level	20		34, 35	16, 37					
F21	f2 (f input), high level	19		34	16, 35, 37					
F22	g2 (g input), high level	17		35	16, 34, 37					
F23	h2 (h input), high level	18		Open	16, 34, 35, 37					
F24	E3 (enable), high level	26			22, 23, 24					
F25	A3 (A control), high level	24			22, 23, 26					
F26	B3 (B control), high level	23			22, 24, 26					
F27	C3 (C control), high level	22		Open	23, 24, 26					
F28	a3 (a input), high level	25		22, 23, 24	26					
F29	b3 (b input), high level	27	Open	22, 23	24, 26	X01-1	0	40	µA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE VIII. Input current, high level, test - Continued.

Test cond.	Parameter	Connect				Read meter	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin		Min	Max		
F30	c3 (c input), high level	28	Open	22, 24	23, 26	X01-1	0	40	µA	10
F31	d3 (d input), high level	29		22	23, 24, 26					
F32	e3 (e input), high level	30		23, 24	22, 26					
F33	f3 (f input), high level	31		23	22, 24, 26					
F34	g3 (g input), high level	32		24	22, 23, 26					
F35	h3 (h input), high level	33	Open	Open	22, 23, 24, 26	X01-1	0	40	µA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE IX. Input current, low level, test.

Test cond.	Parameter	Connect				Read meter	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin		Min	Max		
G1	E1 and E2 (enable pin 37), low level	11, 12, 13, 16, 34, 35	37	Open	Open	X01-2	0.20	3.46	mA	10
G2	A1 (A control), low level	12, 13, 37	11				0.10	1.73		
G3	B1 (B control), low level	11, 13, 37	12							
G4	C1 (C control), low level	11, 12, 37	13		Open					
G5	a1 (a input), low level	11, 12, 13	2		37					
G6	b1 (b input), low level	12, 13	9		11, 37					
G7	c1 (c input), low level	11, 13	8		12, 37					
G8	d1 (d input), low level	13	7		11, 12, 37					
G9	e1 (e input), low level	11, 12	6	Open	13, 37					
G10	f1 (f input), low level	12	5	Open	11, 13, 37					
G11	g1 (g input), low level	11	4		12, 13, 37					
G12	h1 (h input), low level	Open	3		11, 12, 13, 37					
G13	A2 (A control), low level	16, 34, 37	35		Open					
G14	B2 (B control), low level	16, 35, 37	34	Open	Open	X01-2	0.10	1.73	mA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE IX Input current, low level, test - Continued.

Test cond.	Parameter	Connect				Read meter	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin		Min	Max		
G15	C2 (C control), low level	34, 35, 37	16	Open	Open	X01-2	0.10	1.73	mA	10
G16	a2 (a input), low level	16, 34, 35	36		37					
G17	b2 (b input), low level	16, 34	38		35, 37					
G18	c2 (c input), low level	16, 35	39		34, 37					
G19	d2 (d input), low level	16	40		34, 35, 37					
G20	e2 (e input), low level	34, 35	20		16, 37					
G21	f2 (f input), low level	34	19		16, 35, 37					
G22	g2 (g input), low level	35	17		16, 34, 37					
G23	h2 (h input), low level	Open	18		16, 34, 35, 37					
G24	E3 (enable), low level	22, 23, 24	26		Open					
G25	A3 (A control), low level	22, 23, 26	24							
G26	B3 (B control), low level	22, 24, 26	23							
G27	C3 (C control), low level	23, 24, 26	22		Open					
G28	a3 (a input), low level	22, 23, 24	25		26					
G29	b3 (b input), low level	22, 23	27	Open	24, 26	X01-2	0.10	1.73	mA	10

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T15:22Z
Check the source to verify that this is the current version before use.

* TABLE IX. Input current, low level, test - Continued

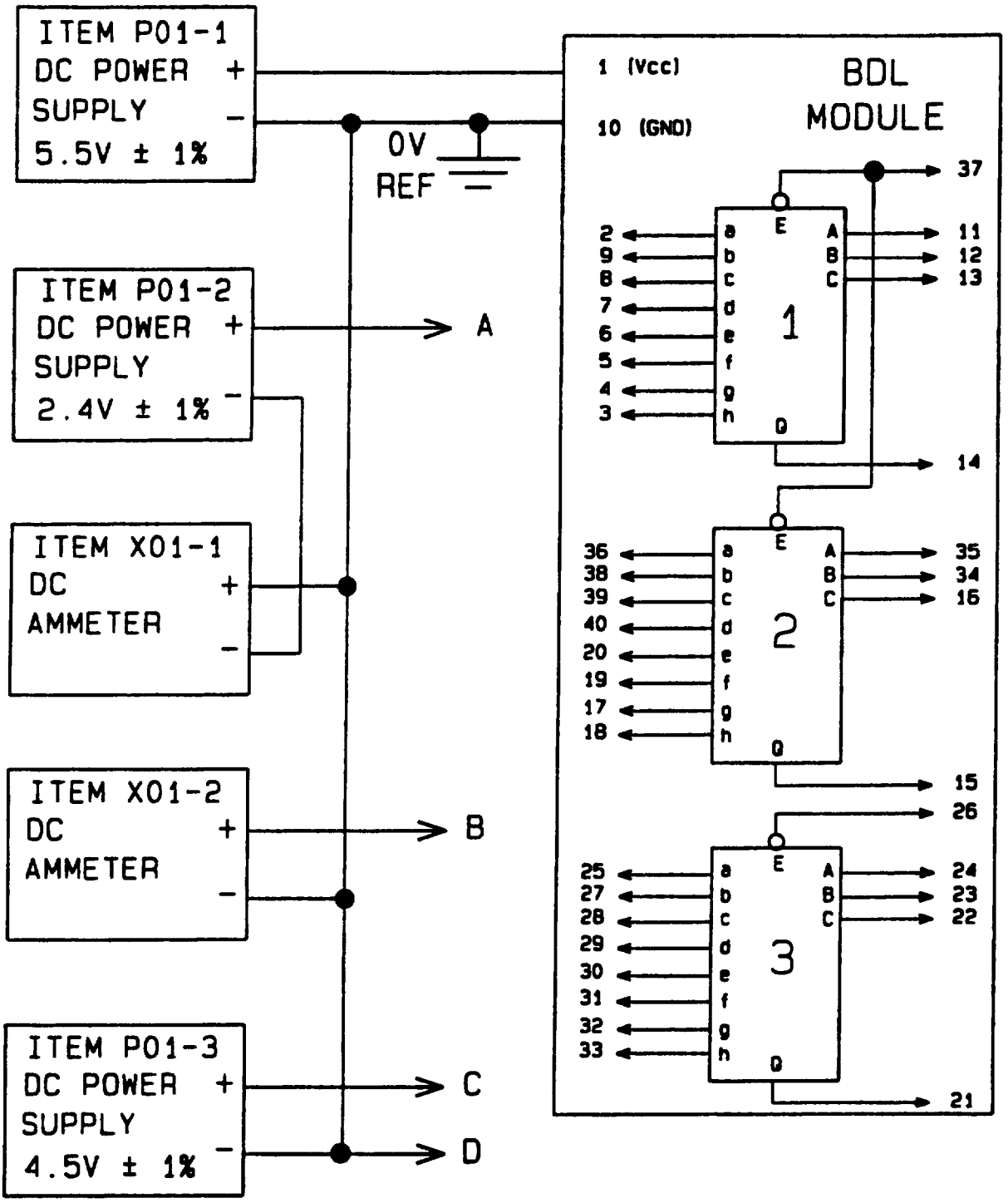
Test cond.	Parameter	Connect				Read meter	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin		Min	Max		
G30	c3 (c input), low level	22, 24	28	Open	23, 26	X01-2	0.10	1.73	mA	10
G31	d3 (d input), low level	22	29		23, 24, 26					
G32	e3 (e input), low level	23, 24	30		22, 26					
G33	f3 (f input), low level	23	31		22, 24, 26					
G34	g3 (g input), low level	24	32		22, 23, 26					
G35	h3 (h input), low level	Open	33	Open	22, 23, 24, 26	X01-2	0.10	1.73	mA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

• TABLE X. Output short circuit test.

Test cond.	Parameter	Connect				Read meter	Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin		Min	Max		
H1	Q1 Output short circuit current	3	14	Open	11, 12, 13, 37	X01-2	20	100	mA	10
H2	Q2 Output short circuit current	18	15		16, 34, 35, 37					
H3	Q3 Output short circuit current	33	21	Open	22, 23, 24, 26	X01-2	20	100	mA	10

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the current version before use.



CAUTION: Do not isolate chassis of ammeter X01 from earth ground.

FIGURE 10. Input current (high level and low level) and output short circuit current test circuit for tables VIII, IX and X.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:22Z
Check the source to verify that this is the current version before use.

NOTES:

1. Since meters X01-1 and X01-2 are not used simultaneously, only one DC ammeter X01 is required.

2. With point "A" disconnected, ammeter X01-1 should indicate 0.0 μA on the 1 μA range.

3. When performing output short circuit current tests, only one output shall be shorted at a time and the measurements taken within 5 seconds.

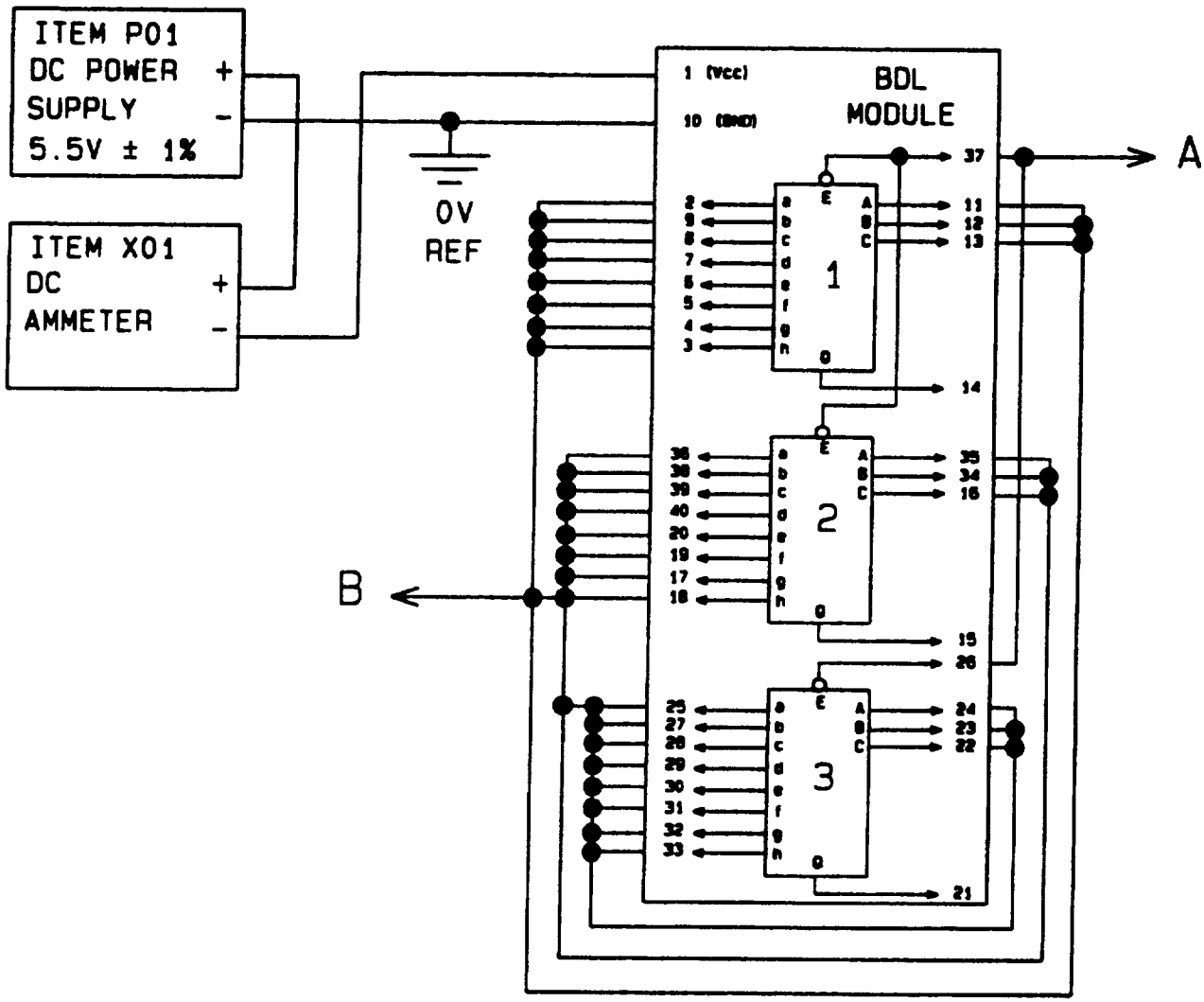
FIGURE 10. Input current (high level and low level) and output short circuit current test circuit for tables VIII, IX and X - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XI. Power supply current (high level and low level) test.

Test cond.	Parameter	Connect Pin 10 to	Initial limits 25° ± 5° Ambient		Unit	Test fig.
			Min	Max		
J1	Power supply current, (low level)	B	48	132	mA	11
J2	Power supply current, (high level)	A	48	132	mA	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



CAUTION: Do not isolate chassis of ammeter X01 from earth ground.

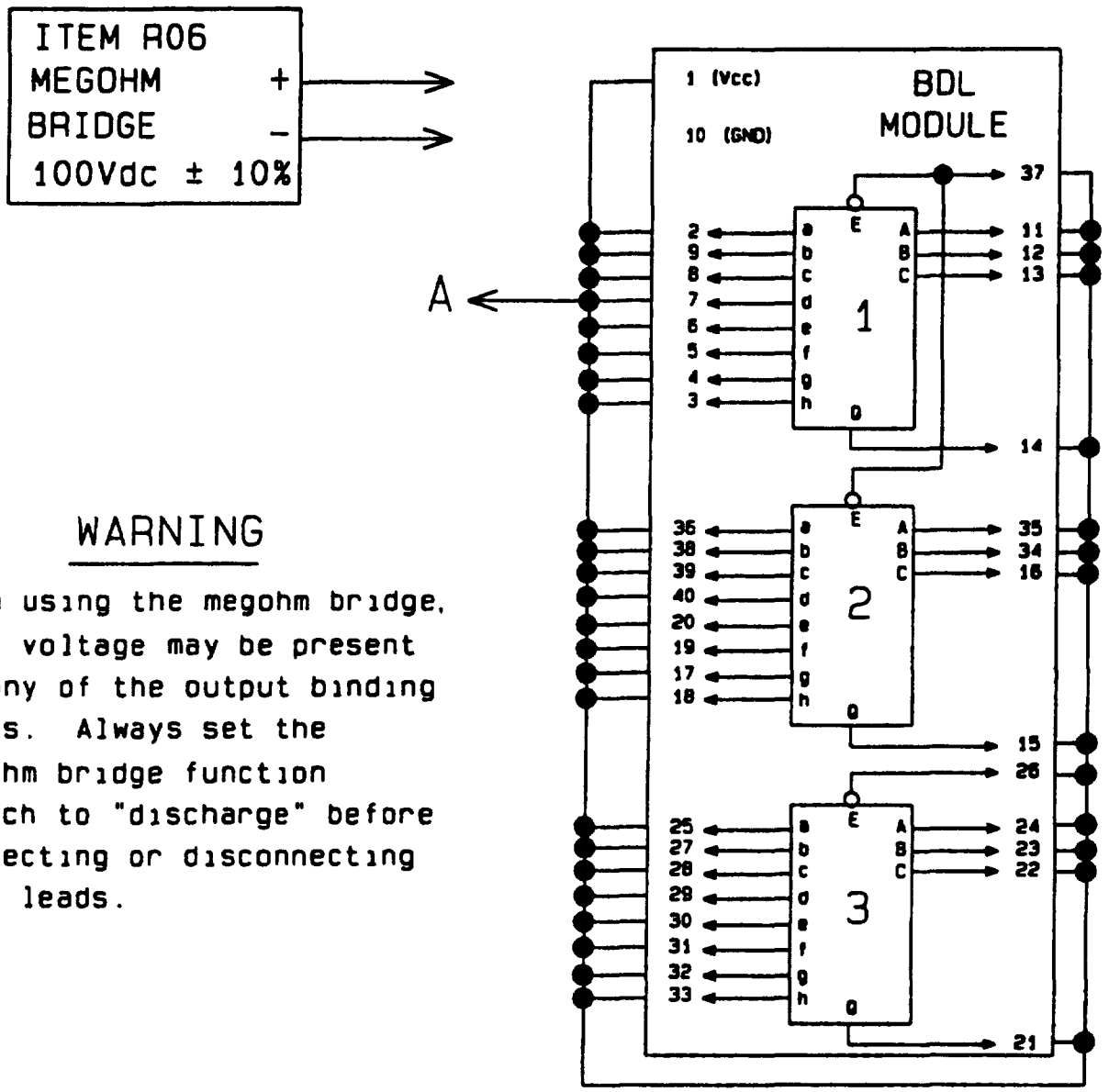
FIGURE 11. Power supply current (high level and low level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XII. Isolation test.

Test cond.	Parameter	Connect		Initial limits 25° ± 5° Ambient		Unit	Test fig.
		Positive to pin	Negative to pin	Min	Max		
K	Isolation	A	Frame	10	INF	MOhm	12

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 12. Isolation test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input capacitance test.

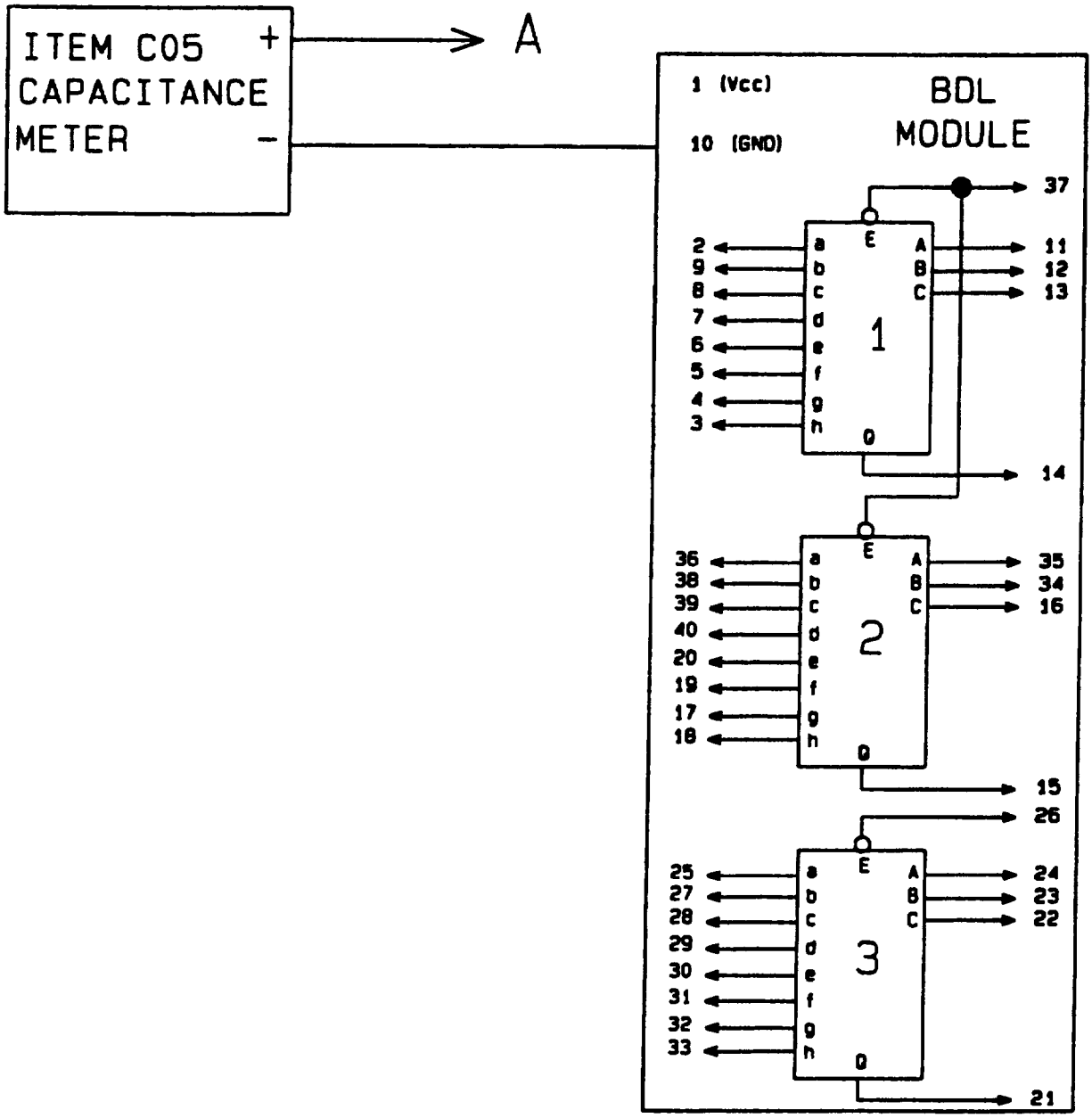
Test cond.	Parameter	Connect A to pin	Initial limits 25° ± 5° C Ambient Min	Initial limits 25° ± 5° C Ambient Max	Unit	Test fig.
L1	E1 and E2 (enable pin 37)	37	4	30	pF	13
L2	a1 (a input)	2	2	15		
L3	b1 (b input)	9				
L4	c1 (c input)	8				
L5	d1 (d input)	7				
L6	e1 (e input)	6				
L7	f1 (f input)	5				
L8	g1 (g input)	4				
L9	h1 (h input)	3				
L10	A1 (A control)	11				
L11	B1 (B control)	12				
L12	C1 (C control)	13				
L13	a2 (a input)	36				
L14	b2 (b input)	38				
L15	c2 (c input)	39				
L16	d2 (d input)	40				
L17	e2 (e input)	20				
L18	f2 (f input)	19				
L19	g2 (g input)	17				
L20	h2 (h input)	18	2	15	pF	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XIII. Input capacitance test - Continued

Test cond.	Parameter	Connect A to pin	Initial limits 25° ± 5° C Ambient		Unit	Test fig.
			Min	Max		
L21	A2 (A control)	35	2	15	pF	13
L22	B2 (B control)	34				
L23	C2 (C control)	16				
L24	E3 (enable)	26				
L25	a3 (a input)	25				
L26	b3 (b input)	27				
L27	c3 (c input)	28				
L28	d3 (d input)	29				
L29	e3 (e input)	30				
L30	f3 (f input)	31				
L31	g3 (g input)	32				
L32	h3 (h input)	33				
L33	A3 (A control)	24				
L34	B3 (B control)	23				
L35	C3 (C control)	22	2	15	pF	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



NOTES:

1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

2. Null out the effects of the test leads prior to connecting to the module.

* FIGURE 13. Input capacitance test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XIV. Transition time (tTLH) test.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
M1	Q1, tTLH	2	14	11, 12, 13	37	5	45	nS	14, 15
M2	Q2, tTLH	36	15	16, 34, 35	37				
M3	Q3, tTLH	25	21	22, 23, 24	26	5	45	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XV. Transition time (tTHL) test.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
N1	Q1, tTHL	2	14	11, 12, 13	37	5	25	nS	14, 15
N2	Q2, tTHL	36	15	16, 34, 35	37				
N3	Q3, tTHL	25	21	22, 23, 24	26	5	25	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P1	Q1, tPLH, a input	2	14	11, 12, 13	37	5	40	nS	14, 15
P2	Q1, tPLH, b input	9		12, 13	11, 37				
P3	Q1, tPLH, c input	8		11, 13	12, 37				
P4	Q1, tPLH, d input	7		13	11, 12, 37				
P5	Q1, tPLH, e input	6		11, 12	13, 37				
P6	Q1, tPLH, f input	5		12	11, 13, 37				
P7	Q1, tPLH, g input	4		11	12, 13, 37				
P8	Q1, tPLH, h input	3	14	Open	11, 12, 13, 37				
P9	Q2, tPLH, a input	36	15	16, 34, 35	37				
P10	Q2, tPLH, b input	38		16, 34	35, 37				
P11	Q2, tPLH, c input	39		16, 35	34, 37				
P12	Q2, tPLH, d input	40		16	34, 35, 37				
P13	Q2, tPLH, e input	20	15	34, 35	16, 37	5	40	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P14	Q2, tPLH, f input	19	15	34	16, 35, 37	5	40	nS	14, 15
P15	Q2, tPLH, g input	17		35	16, 34, 37				
P16	Q2, tPLH, h input	18	15	Open	16, 34, 35, 37				
P17	Q3, tPLH, a input	25	21	22, 23, 24	26				
P18	Q3, tPLH, b input	27		22, 23	24, 26				
P19	Q3, tPLH, c input	28		22, 24	23, 26				
P20	Q3, tPLH, d input	29		22	23, 24, 26				
P21	Q3, tPLH, e input	30		23, 24	22, 26				
P22	Q3, tPLH, f input	31		23	22, 24, 26				
P23	Q3, tPLH, g input	32		24	22, 23, 26				
P24	Q3, tPLH, h input	33	21	Open	22, 23, 24, 26	5	40		14, 15
P25	Q1, tPLH, enable a	37	14	2, 11, 12, 13	Open	7	62	nS	14, 18

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P26	Q1, tPLH, enable b	37	14	9, 12, 13	11	7	62	nS	14, 16
P27	Q1, tPLH, enable c			8, 11, 13	12				
P28	Q1, tPLH, enable d			7, 13	11, 12				
P29	Q1, tPLH, enable e			6, 11, 12	13				
P30	Q1, tPLH, enable f			5, 12	11, 13				
P31	Q1, tPLH, enable g			4, 11	12, 13				
P32	Q1, tPLH, enable h		14	3	11, 12, 13				
P33	Q2, tPLH, enable a		15	16, 34, 35, 38	Open				
P34	Q2, tPLH, enable b			16, 34, 38	35				
P35	Q2, tPLH, enable c			16, 35, 39	34				
P36	Q2, tPLH, enable d			16, 40	34, 35				
P37	Q2, tPLH, enable e			20, 34, 35	16				
P38	Q2, tPLH, enable f			19, 34	16, 35				
P39	Q2, tPLH, enable g	37	15	17, 35	16, 34	7	62	nS	14, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P40	Q2, tPLH, enable h	37	15	18	16, 34, 35	7	62	nS	14, 16
P41	Q3, tPLH, enable a	26	21	22, 23, 24, 25	Open				
P42	Q3, tPLH, enable b			22, 23, 27	24				
P43	Q3, tPLH, enable c			22, 24, 28	23				
P44	Q3, tPLH, enable d			22, 29	23, 24				
P45	Q3, tPLH, enable e			23, 24, 30	22				
P46	Q3, tPLH, enable f			23, 31	22, 24				
P47	Q3, tPLH, enable g			24, 32	22, 23				
P48	Q3, tPLH, enable h	26	21	33	22, 23, 24				14, 16
P49	Q1, tPLH, A control	11	14	2, 12, 13	9, 37				14, 15
P50	Q1, tPLH, A control			9, 12, 13	2, 37				14, 16
P51	Q1, tPLH, A control			8, 13	7, 12, 37				14, 15
P52	Q1, tPLH, A control	11	14	7, 13	8, 12, 37	7	62	nS	14, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P53	Q1, tPLH, A control	11	14	6, 12	5, 13, 37	7	62	nS	14, 15
P54	Q1, tPLH, A control			5, 12	6, 13, 37				14, 16
P55	Q1, tPLH, A control			4	3, 12, 13, 37				14, 15
P56	Q1, tPLH, A control	11		3	4, 12, 13, 37				14, 16
P57	Q1, tPLH, B control	12		2, 11, 13	8, 37				14, 15
P58	Q1, tPLH, B control			8, 11, 13	2, 37				14, 16
P59	Q1, tPLH, B control			9, 13	7, 11, 37				14, 15
P60	Q1, tPLH, B control			7, 13	9, 11, 37				14, 16
P61	Q1, tPLH, B control			6, 11	4, 13, 37				14, 15
P62	Q1, tPLH, B control			4, 11	6, 13, 37				14, 16
P63	Q1, tPLH, B control			5	3, 11, 13, 37				14, 15
P64	Q1, tPLH, B control	12	14	3	5, 11, 13, 37	7	62	nS	14, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P65	Q1, tPLH, C control	13	14	2, 11, 12	6, 37	7	62	nS	14, 15
P66	Q1, tPLH, C control			6, 11, 12	2, 37				14, 16
P67	Q1, tPLH, C control			9, 12	5, 11, 37				14, 15
P68	Q1, tPLH, C control			5, 12	9, 11, 37				14, 16
P69	Q1, tPLH, C control	13		8, 11	4, 12, 37				14, 15
P70	Q1, tPLH, C control	13		4, 11	8, 12, 37				14, 16
P71	Q1, tPLH, C control			7	3, 11, 12, 37				14, 15
P72	Q1, tPLH, C control	13	14	3	7, 11, 12, 37				14, 16
P73	Q2, tPLH, A control	35	15	16, 34, 36	37, 38				14, 15
P74	Q2, tPLH, A control			16, 34, 38	36, 37				14, 16
P75	Q2, tPLH, A control	35	15	16, 39	34, 37, 40	7	62	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P76	Q2, tPLH, A control	35	15	16, 40	34, 37, 39	7	62	nS	14, 16
P77	Q2, tPLH, A control			20, 34	16, 19, 37				14, 15
P78	Q2, tPLH, A control			19, 34	16, 20, 37				14, 16
P79	Q2, tPLH, A control			17	16, 18, 34, 37				14, 15
P80	Q2, tPLH, A control	35		18	16, 17, 34, 37				14, 16
P81	Q2, tPLH, B control	34		16, 35, 36	37, 39				14, 15
P82	Q2, tPLH, B control			16, 35, 39	36, 37				14, 16
P83	Q2, tPLH, B control			16, 38	35, 37, 40				14, 15
P84	Q2, tPLH, B control			16, 40	35, 37, 38				14, 16
P85	Q2, tPLH, B control			20, 35	16, 17, 37				14, 15
P86	Q2, tPLH, B control			17, 35	16, 20, 37				14, 16
P87	Q2, tPLH, B control	34	15	19	16, 18, 35, 37	7	62	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P88	Q2, tPLH, B control	34	15	18	16, 19, 35, 37	7	62	nS	14, 16
P89	Q2, tPLH, C control	16		34, 35, 36	20, 37				14, 15
P90	Q2, tPLH, C control			20, 34, 35	36, 37				14, 16
P91	Q2, tPLH, C control			34, 38	19, 35, 37				14, 15
P92	Q2, tPLH, C control			19, 34	35, 37, 38				14, 16
P93	Q2, tPLH, C control			35, 39	17, 34, 37				14, 15
P94	Q2, tPLH, C control			17, 35	34, 37, 39				14, 16
P95	Q2, tPLH, C control			40	18, 34, 35, 37				14, 15
P96	Q2, tPLH, C control	16	15	18	34, 35, 37, 40				14, 16
P97	Q3, tPLH, A control	24	21	22, 23, 25	26, 27				14, 15
P98	Q3, tPLH, A control			22, 23, 27	25, 26				14, 16
P99	Q3, tPLH, A control	24	21	22, 28	23, 26, 29	7	62	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P100	Q3, tPLH, A control	24	21	22, 29	23, 26, 28	7	62	nS	14, 16
P101	Q3, tPLH, A control			23, 30	22, 26, 30				14, 15
P102	Q3, tPLH, A control			23, 31	22, 26, 30				14, 16
P103	Q3, tPLH, A control			32	22, 23, 26, 33				14, 15
P104	Q3, tPLH, A control	24		33	22, 23, 26, 32				14, 16
P105	Q3, tPLH, B control	23		22, 24, 25	26, 28				14, 15
P106	Q3, tPLH, B control			22, 24, 28	25, 26				14, 16
P107	Q3, tPLH, B control			22, 27	24, 26, 29				14, 15
P108	Q3, tPLH, B control			22, 29	24, 26, 27				14, 16
P109	Q3, tPLH, B control			24, 30	22, 26, 32				14, 15
P110	Q3, tPLH, B control			24, 32	22, 26, 30				14, 16
P111	Q3, tPLH, B control	23	21	31	22, 24, 26, 33	7	62	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (tPLH) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
P112	Q3, tPLH, B control	23	21	33	22, 24, 26, 31	7	62	nS	14, 16
P113	Q3, tPLH, C control	22		23, 24, 25	26, 30				14, 15
P114	Q3, tPLH, C control			23, 24, 30	25, 26				14, 16
P115	Q3, tPLH, C control			23, 27	24, 26, 31				14, 15
P116	Q3, tPLH, C control			23, 31	24, 26, 27				14, 16
P117	Q3, tPLH, C control			24, 28	23, 26, 32				14, 15
P118	Q3, tPLH, C control			24, 32	23, 26, 28				14, 16
P119	Q3, tPLH, C control			29	23, 24, 26, 33				14, 15
P120	Q3, tPLH, C control	22	21	33	23, 24, 26, 29	7	62		14, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII Propagation delay time (tPHL) test.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q1	Q1, tPHL, a input	2	14	11, 12, 13	37	5	40	nS	14, 15
Q2	Q1, tPHL, b input	9		12, 13	11, 37				
Q3	Q1, tPHL, c input	8		11, 13	12, 37				
Q4	Q1, tPHL, d input	7		13	11, 12, 37				
Q5	Q1, tPHL, e input	6		11, 12	13, 37				
Q6	Q1, tPHL, f input	5		12	11, 13, 37				
Q7	Q1, tPHL, g input	4		11	12, 13, 37				
Q8	Q1, tPHL, h input	3	14	Open	11, 12, 13, 37				
Q9	Q2, tPHL, a input	36	15	16, 34, 35	37				
Q10	Q2, tPHL, b input	38		16, 34	35, 37				
Q11	Q2, tPHL, c input	39		16, 35	34, 37				
Q12	Q2, tPHL, d input	40		16	34, 35, 37				
Q13	Q2, tPHL, e input	20	15	34, 35	16, 37	5	40	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q14	Q2, tPHL, f input	19	15	34	16, 35, 37	5	40	nS	14, 15
Q15	Q2, tPHL, g input	17		35	16, 34, 37				
Q16	Q2, tPHL, h input	18	15	Open	16, 34, 35, 37				
Q17	Q3, tPHL, a input	25	21	22, 23, 24	26				
Q18	Q3, tPHL, b input	27		22, 23	24, 26				
Q19	Q3, tPHL, c input	28		22, 24	23, 26				
Q20	Q3, tPHL, d input	29		22	23, 24, 26				
Q21	Q3, tPHL, e input	30		23, 24	22, 26				
Q22	Q3, tPHL, f input	31		23	22, 24, 26				
Q23	Q3, tPHL, g input	32		24	22, 23, 26				14, 15
Q24	Q3, tPHL, h input	33	21	Open	22, 23, 24, 26	5	40		14, 15
Q25	Q1, tPHL, enable a	37	14	2, 11, 12, 13	Open	7	50	nS	14, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q26	Q1, tPHL, enable b	37	14	9, 12, 13	11	7	50	nS	14, 16
Q27	Q1, tPHL, enable c			8, 11, 13	12				
Q28	Q1, tPHL, enable d			7, 13	11, 12				
Q29	Q1, tPHL, enable e			6, 11, 12	13				
Q30	Q1, tPHL, enable f			5, 12	11, 13				
Q31	Q1, tPHL, enable g			4, 11	12, 13				
Q32	Q1, tPHL, enable h		14	3	11, 12, 13				
Q33	Q2, tPHL, enable a		15	16, 34, 35, 36	Open				
Q34	Q2, tPHL, enable b			16, 34, 38	35				
Q35	Q2, tPHL, enable c			16, 35, 39	34				
Q36	Q2, tPHL, enable d			16, 40	34, 35				
Q37	Q2, tPHL, enable e			20, 34, 35	16				
Q38	Q2, tPHL, enable f	37	15	19, 34	16, 35	7	50	nS	14, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q39	Q2, tPHL, enable g	37	15	17, 35	16, 34	7	50	nS	14, 16
Q40	Q2, tPHL, enable h	37	15	18	16, 34, 35				
Q41	Q3, tPHL, enable a	26	21	22, 23, 24, 25	Open				
Q42	Q3, tPHL, enable b			22, 23, 27	24				
Q43	Q3, tPHL, enable c			22, 24, 28	23				
Q44	Q3, tPHL, enable d			22, 29	23, 24				
Q45	Q3, tPHL, enable e			23, 24, 30	22				
Q46	Q3, tPHL, enable f			23, 31	22, 24				
Q47	Q3, tPHL, enable g			24, 32	22, 23				
Q48	Q3, tPHL, enable h	26	21	33	22, 23, 24				14, 16
Q49	Q1, tPHL, A control	11	14	2, 12, 13	9, 37				14, 15
Q50	Q1, tPHL, A control			9, 12, 13	2, 37				14, 16
Q51	Q1, tPHL, A control	11	14	8, 13	7, 12, 37	7	50	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q52	Q1, tPHL, A control	11	14	7, 13	8, 12, 37	7	50	nS	14, 16
Q53	Q1, tPHL, A control			6, 12	5, 13, 37				14, 15
Q54	Q1, tPHL, A control			5, 12	6, 13, 37				14, 16
Q55	Q1, tPHL, A control			4	3, 12, 13, 37				14, 15
Q56	Q1, tPHL, A control	11		3	4, 12, 13, 37				14, 16
Q57	Q1, tPHL, B control	12		2, 11, 13	8, 37				14, 15
Q58	Q1, tPHL, B control			8, 11, 13	2, 37				14, 16
Q59	Q1, tPHL, B control			9, 13	7, 11, 37				14, 15
Q60	Q1, tPHL, B control			7, 13	9, 11, 37				14, 16
Q61	Q1, tPHL, B control			6, 11	4, 13, 37				14, 15
Q62	Q1, tPHL, B control			4, 11	6, 13, 37				14, 16
Q63	Q1, tPHL, B control	12	14	5	3, 11, 13, 37	7	50	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q64	Q1, tPHL, B control	12	14	3	5, 11, 13, 37	7	50	nS	14, 16
Q65	Q1, tPHL, C control	13		2, 11, 12	6, 37				14, 15
Q66	Q1, tPHL, C control			6, 11, 12	2, 37				14, 16
Q67	Q1, tPHL, C control			9, 12	5, 11, 37				14, 15
Q68	Q1, tPHL, C control			5, 12	9, 11, 37				14, 16
Q69	Q1, tPHL, C control			8, 11	4, 12, 37				14, 15
Q70	Q1, tPHL, C control			4, 11	8, 12, 37				14, 16
Q71	Q1, tPHL, C control			7	3, 11, 12, 37				14, 15
Q72	Q1, tPHL, C control	13	14	3	7, 11, 12, 37				14, 16
Q73	Q2, tPHL, A control	35	15	16, 34, 36	37, 38				14, 15
Q74	Q2, tPHL, A control			16, 34, 38	36, 37				14, 16
Q75	Q2, tPHL, A control	35	15	16, 39	34, 37, 40	7	50	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q76	Q2, tPHL, A control	35	15	16, 40	34, 37, 39	7	50	nS	14, 16
Q77	Q2, tPHL, A control			20, 34	16, 19, 37				14, 15
Q78	Q2, tPHL, A control			19, 34	16, 20, 37				14, 16
Q79	Q2, tPHL, A control			17	16, 18, 34, 37				14, 15
Q80	Q2, tPHL, A control	35		18	16, 17, 34, 37				14, 16
Q81	Q2, tPHL, B control	34		16, 35, 38	37, 39				14, 15
Q82	Q2, tPHL, B control			16, 35, 39	36, 37				14, 16
Q83	Q2, tPHL, B control			16, 38	35, 37, 40				14, 15
Q84	Q2, tPHL, B control			16, 40	35, 37, 38				14, 16
Q85	Q2, tPHL, B control			20, 35	16, 17, 37				14, 15
Q86	Q2, tPHL, B control			17, 35	16, 20, 37				14, 16
Q87	Q2, tPHL, B control	34	15	19	16, 18, 35, 37	7	50	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q88	Q2, tPHL, B control	34	15	18	16, 19, 35, 37	7	50	nS	14, 16
Q89	Q2, tPHL, C control	16		34, 35, 36	20, 37				14, 15
Q90	Q2, tPHL, C control			20, 34, 35	36, 37				14, 16
Q91	Q2, tPHL, C control			34, 38	19, 35, 37				14, 15
Q92	Q2, tPHL, C control			19, 34	35, 37, 38				14, 16
Q93	Q2, tPHL, C control			35, 39	17, 34, 37				14, 15
Q94	Q2, tPHL, C control			17, 35	34, 37, 39				14, 16
Q95	Q2, tPHL, C control			40	18, 34, 35, 37				14, 15
Q96	Q2, tPHL, C control	16	15	18	34, 35, 37, 40				14, 16
Q97	Q3, tPHL, A control	24	21	22, 23, 25	26, 27				14, 15
Q98	Q3, tPHL, A control			22, 23, 27	25, 26				14, 16
Q99	Q3, tPHL, A control	24	21	22, 28	23, 26, 29	7	50	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

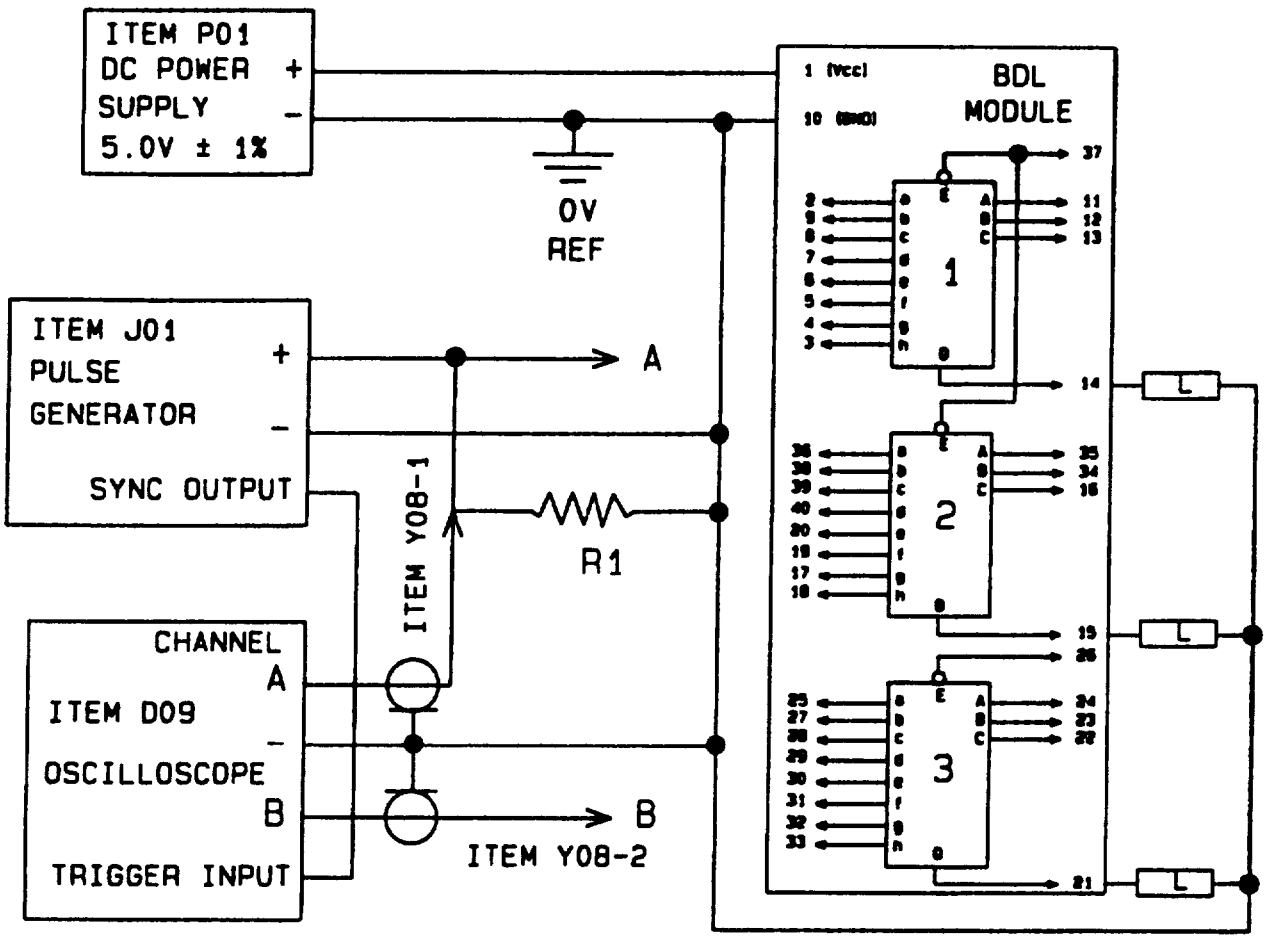
Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q100	Q3, tPHL, A control	24	21	22, 29	23, 26, 28	7	50	nS	14, 16
Q101	Q3, tPHL, A control			23, 30	22, 26, 31				14, 15
Q102	Q3, tPHL, A control			23, 31	22, 26, 30				14, 16
Q103	Q3, tPHL, A control			32	22, 23, 26, 33				14, 15
Q104	Q3, tPHL, A control	24		33	22, 23, 26, 32				14, 16
Q105	Q3, tPHL, B control	23		22, 24, 25	26, 28				14, 15
Q106	Q3, tPHL, B control			22, 24, 28	25, 28				14, 16
Q107	Q3, tPHL, B control			22, 27	24, 26, 29				14, 15
Q108	Q3, tPHL, B control			22, 29	24, 26, 27				14, 16
Q109	Q3, tPHL, B control			24, 30	22, 26, 32				14, 15
Q110	Q3, tPHL, B control			24, 32	22, 26, 30				14, 16
Q111	Q3, tPHL, B control	23	21	31	22, 24, 26, 33	7	50	nS	14, 15

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (tPHL) test - Continued.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	Pin 1 to pin	Pin 10 to pin	Min	Max		
Q112	Q3, tPHL, B control	23	21	33	22, 24, 26, 31	7	50	nS	14, 16
Q113	Q3, tPHL, C control	22		23, 24, 25	26, 30				14, 15
Q114	Q3, tPHL, C control			23, 24, 30	25, 26				14, 16
Q115	Q3, tPHL, C control			23, 27	24, 31, 38				14, 15
Q116	Q3, tPHL, C control			23, 31	24, 26, 27				14, 16
Q117	Q3, tPHL, C control			24, 28	23, 26, 32				14, 15
Q118	Q3, tPHL, C control			24, 32	23, 26, 28				14, 16
Q119	Q3, tPHL, C control			29	23, 24, 26, 33				14, 15
Q120	Q3, tPHL, C control	22	21	33	23, 24, 26, 29	7	50	nS	14, 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



NOTES:

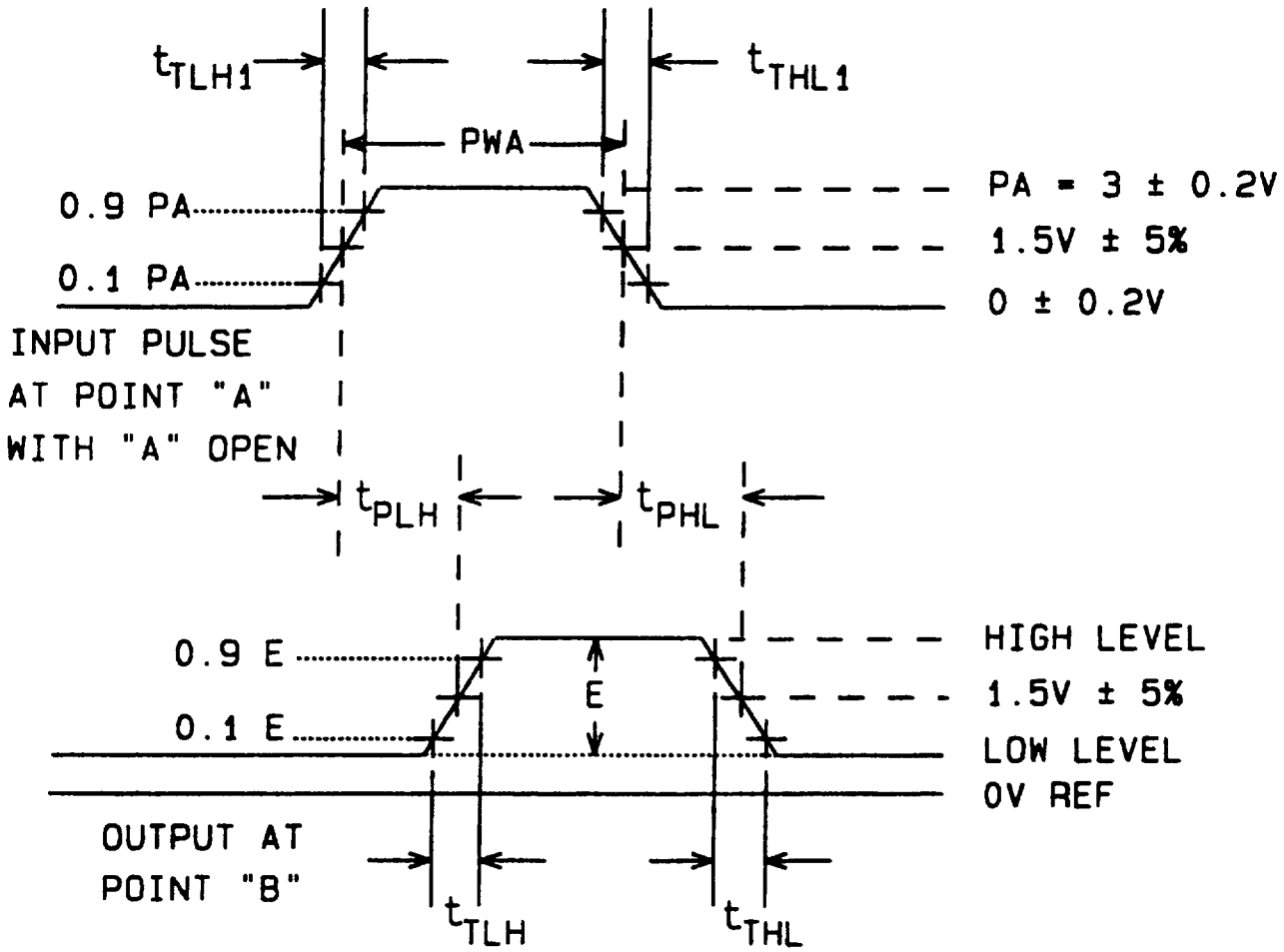
1. Load L is specified on figure 9. Only the output pin under test is required to be loaded.

2. Before making the output pulse characteristics measurements, the output levels shall be verified to be 0.4 V maximum for the low level and 2.4 V minimum for high level,

3. R1 = 50 Ohms ± 5 percent, 1 W.

* FIGURE 14. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for tables XIV, XV, XVI and XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



NOTE: Input parameters:

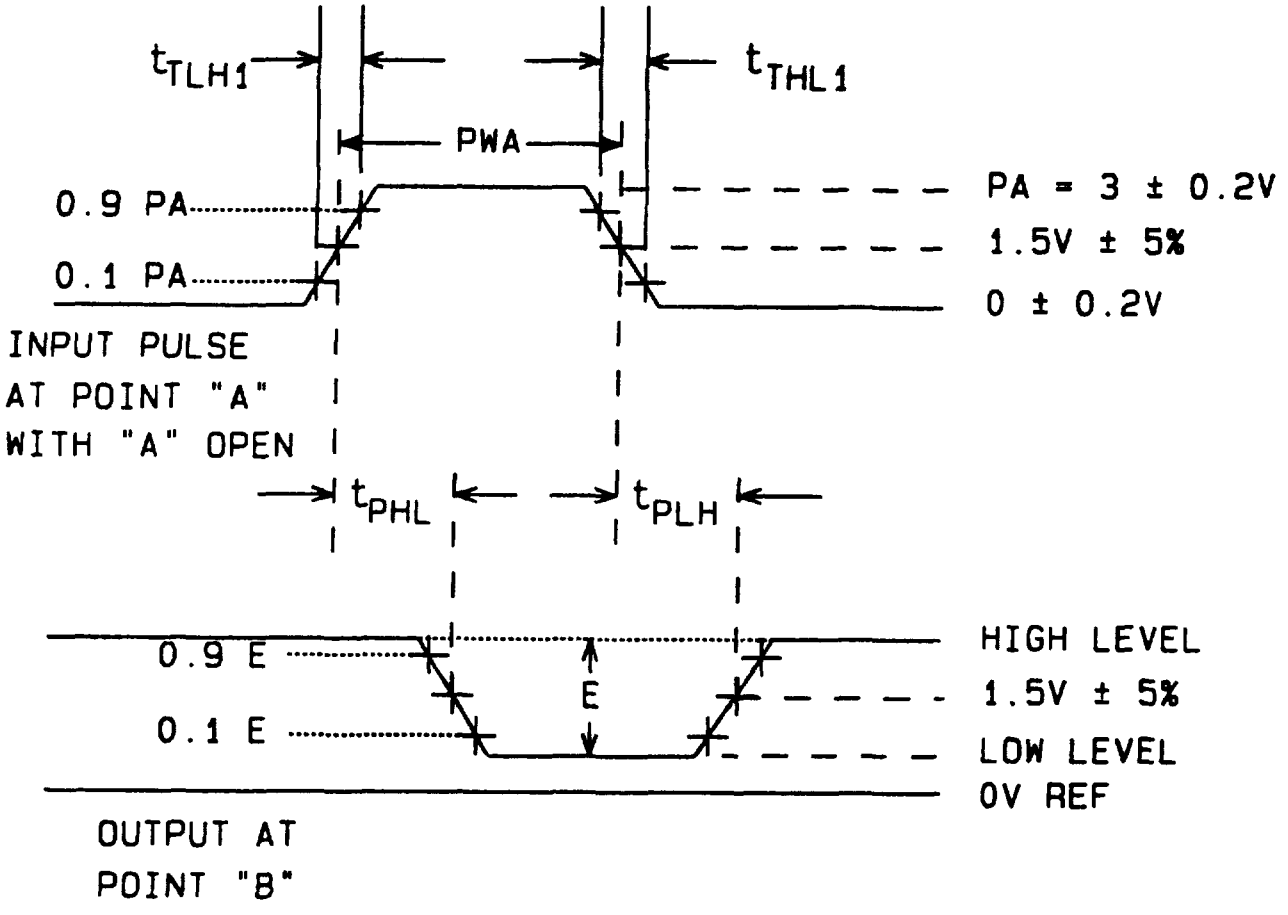
PWA = 200 ± 20 nS

$t_{TLH1} = t_{THL1} = 6 \pm 1$ nS

PRP = 1 ± 0.1 µS

FIGURE 15. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit waveforms for tables XIV, XV, XVI and XVII test conditions M1-M3, N1-N3, P1-P24, P49-P120 odd, Q1-Q24 and Q49-Q120 odd.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the currentversion before use.



NOTE: Input parameters:

PWA = 200 ± 20 nS

$t_{TLH1} = t_{THL1} = 6 \pm 1$ nS

PRP = 1 ± 0.1 µS

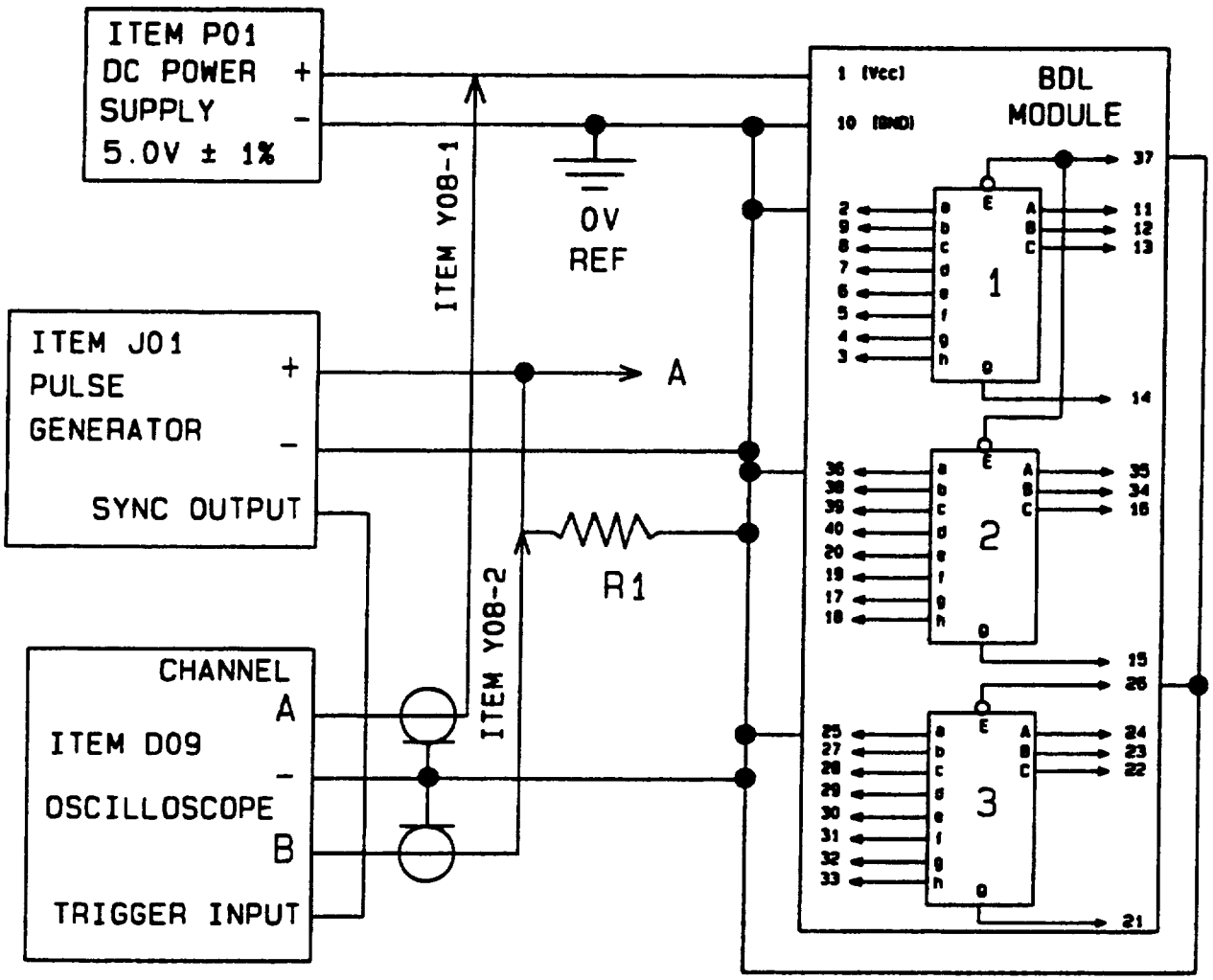
FIGURE 16. Propagation delay time (tPLH and tPHL) test circuit waveforms for tables XVI and XVII test conditions P25-P48, Q25- Q48, P49-P120 even and Q49-Q120 even.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Power supply transient amplitute test.

Test cond.	Parameter	Connect	Initial limits 25° ± 5° Ambient		Unit	Test figure
		A to pin	Min	Max		
R	PSTA	11, 24, 35	0	400	mV	17, 18

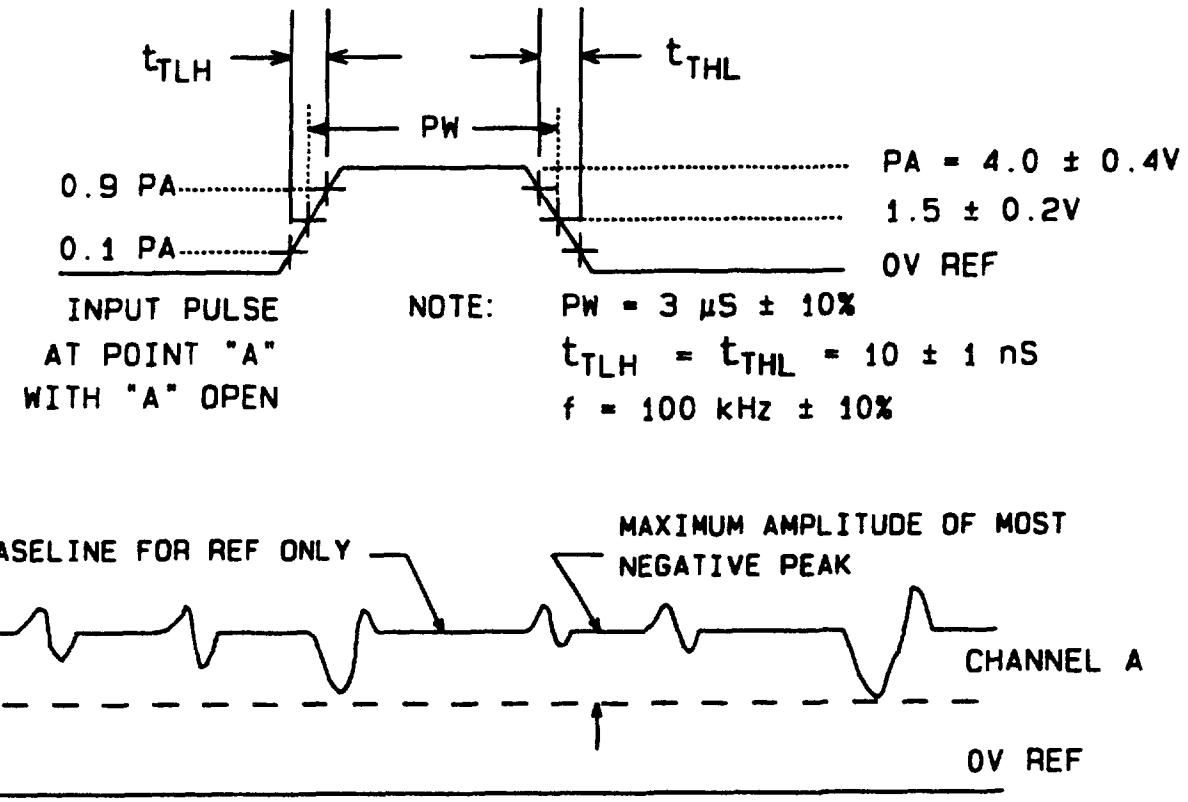
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 50 Ohms ± 5 percent, 1 W.

* FIGURE 17. Power supply transient amplitude test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.



* FIGURE 18. Power supply transient amplitude test circuit waveforms for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XIX. Output crosstalk (high level) test.

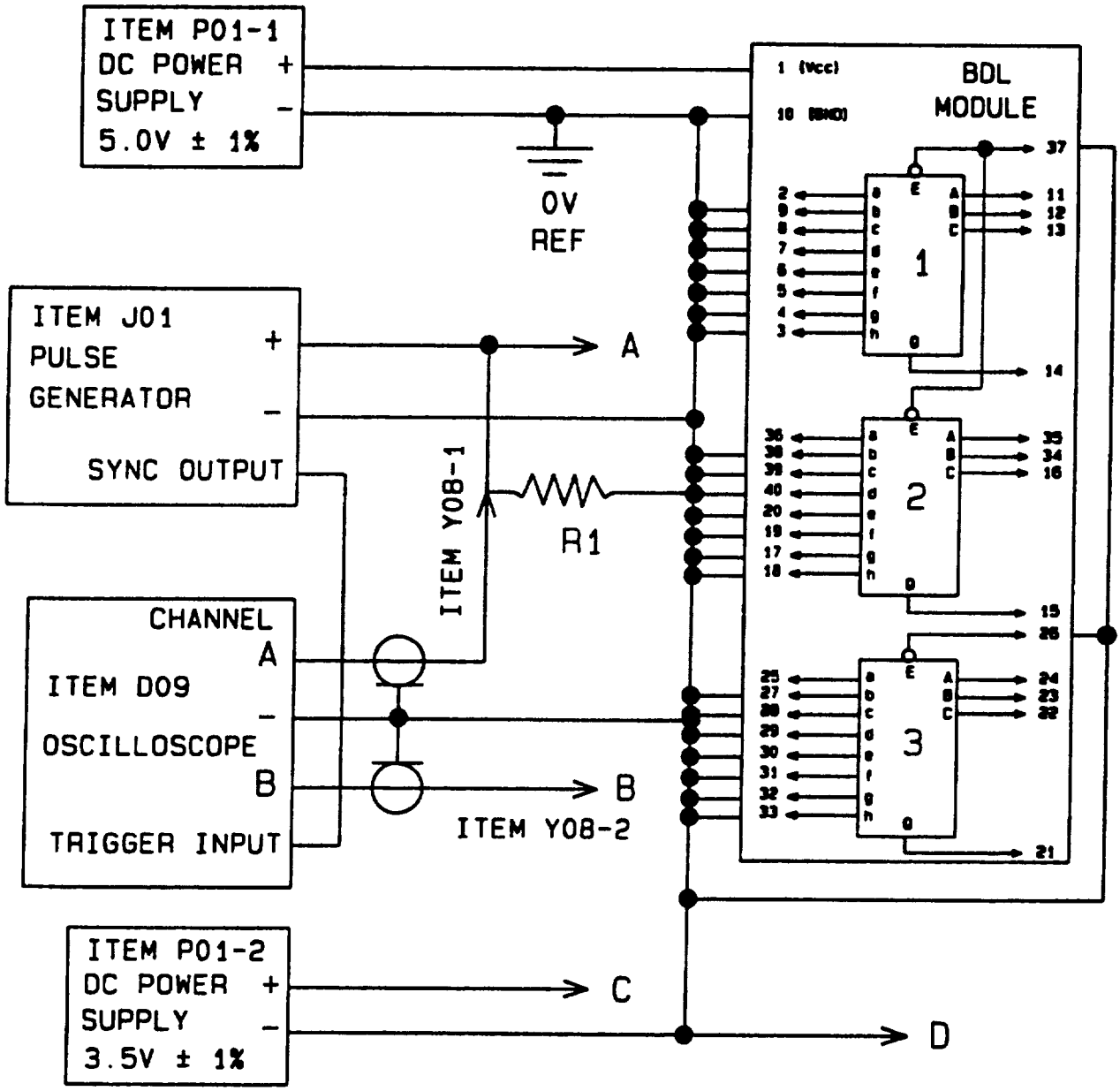
Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	Q1, high level	25, 36	14	2, 11, 12, 13, 16, 22, 23, 24, 34, 35	Open	0	400	mV	19, 20
S2	Q2, high level	2, 25	15	11, 12, 13, 16, 22, 23, 24, 34, 35, 36					
S3	Q3, high level	2, 36	21	11, 12, 13, 16, 22, 23, 24, 25, 34, 35	Open	0	400	mV	19, 20

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XX. Output crosstalk (low level) test.

Test cond.	Parameter	Connect				Initial limits 25° ± 5°C Ambient		Unit	Test fig.
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	Q1, low level	25, 36	14	11, 12, 13, 16, 22, 23, 24, 34, 35	2	0	400	mV	19, 20
T2	Q2, low level	2, 25	15	11, 12, 13, 16, 22, 23, 24, 34, 35	36				
T3	Q3, low level	2, 36	21	11, 12, 13, 16, 22, 23, 24, 34, 35	25	0	400	mV	19, 20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

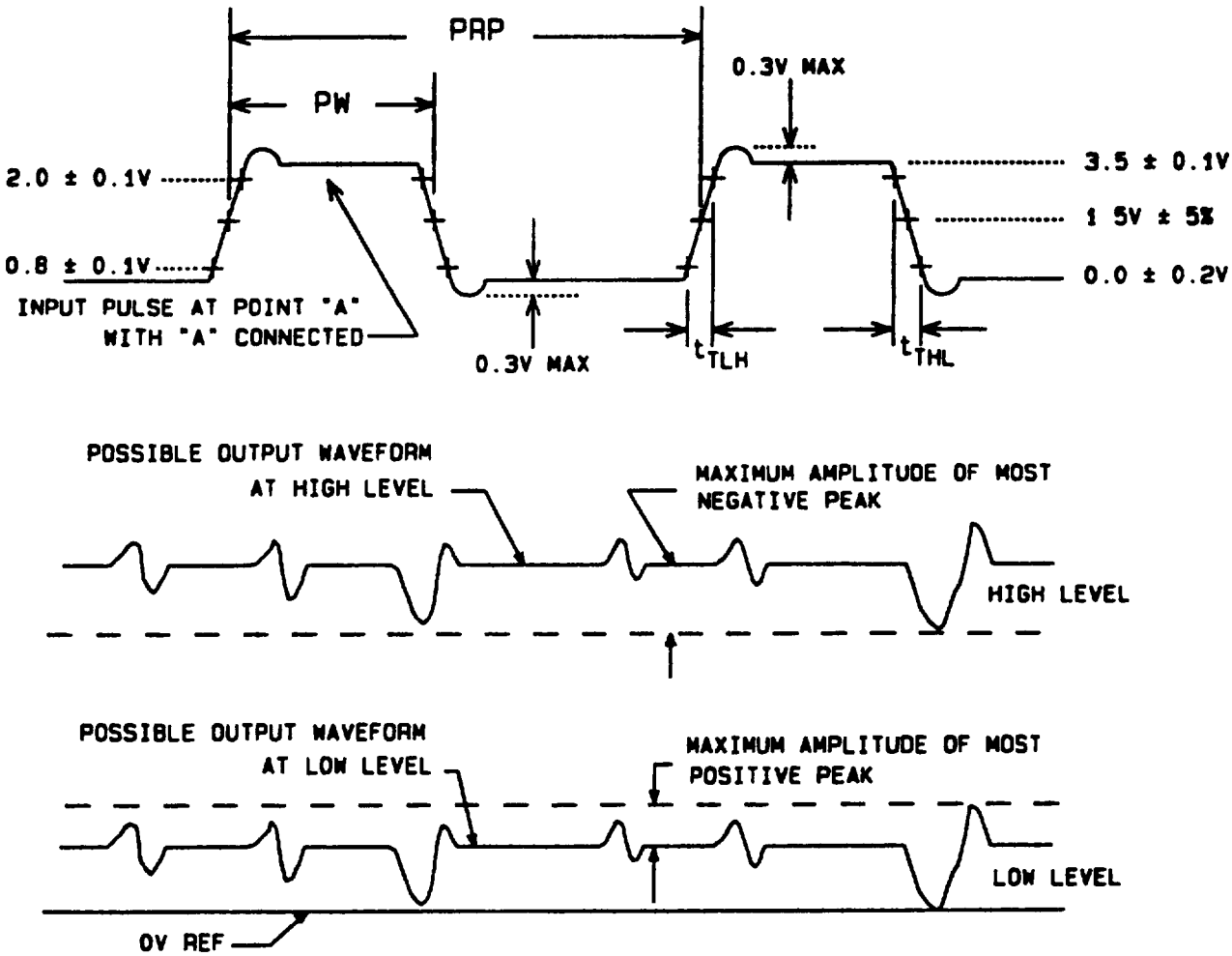


NOTES:

1. R1 = 50 Ohms ± 5 percent, 1 W.
2. Load each output with the load specified on figure 9.

* FIGURE 19. Output crosstalk (high level and low level) test circuit for tables XIX and XX.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:22Z
Check the source to verify that this is the current version before use.



NOTE: Input pulse parameters:

$t_{TLH} = t_{THL}$ = 3 5 ± 1.5 nS (verified at each input pin)

PW = 200 ± 20 nS

PRP = 500 ± 50 nS

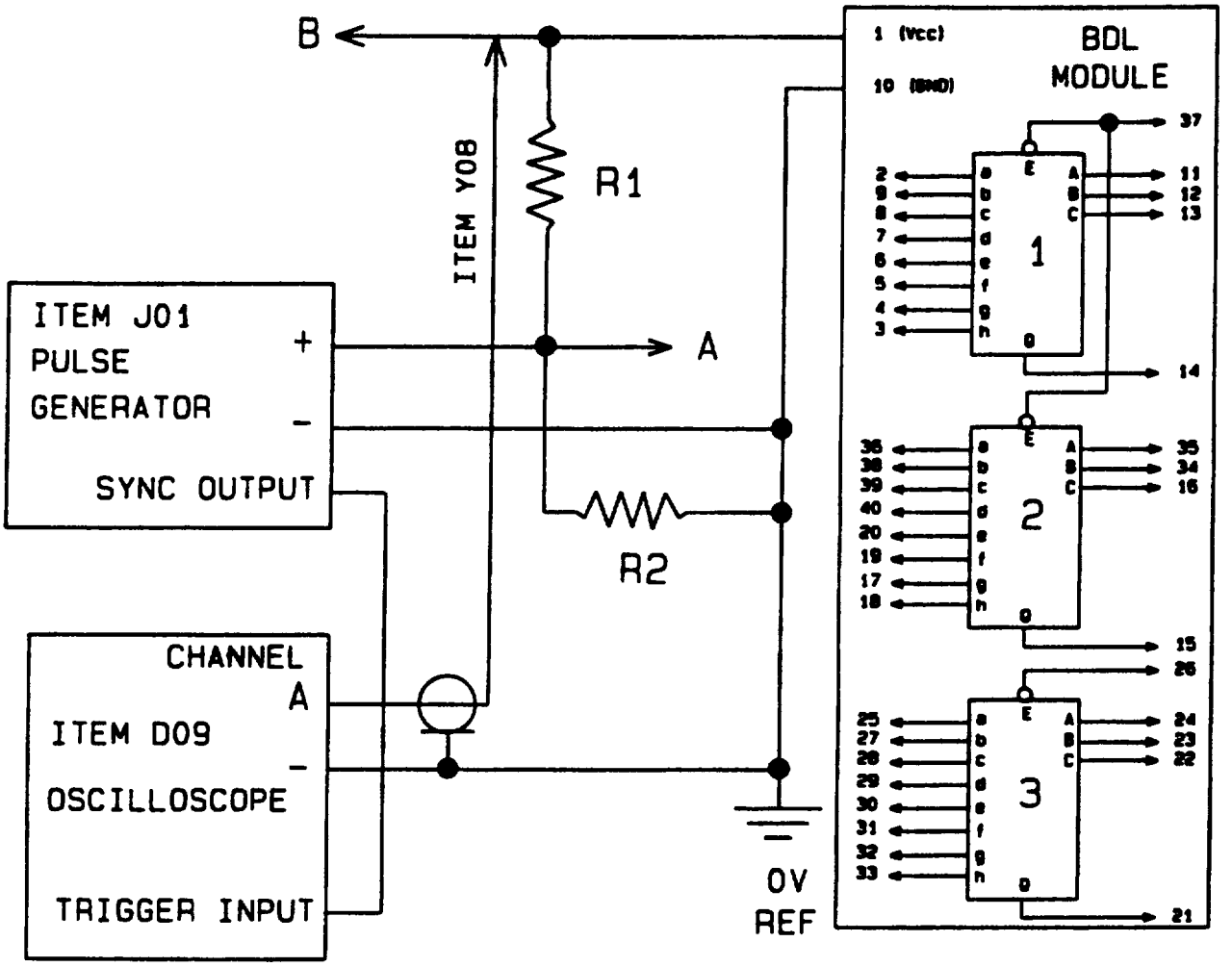
FIGURE 20. Output crosstalk (high level and low level) test circuit waveforms for tables XIX and XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.

TABLE XXI. Power supply transient rise time test.

Test cond.	Parameter	Initial limits 25° ± 5° Ambient		Unit	Test fig.
		Min	Max		
U	Transient rise time	2.5	7.5	μS	21, 22

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the current version before use.

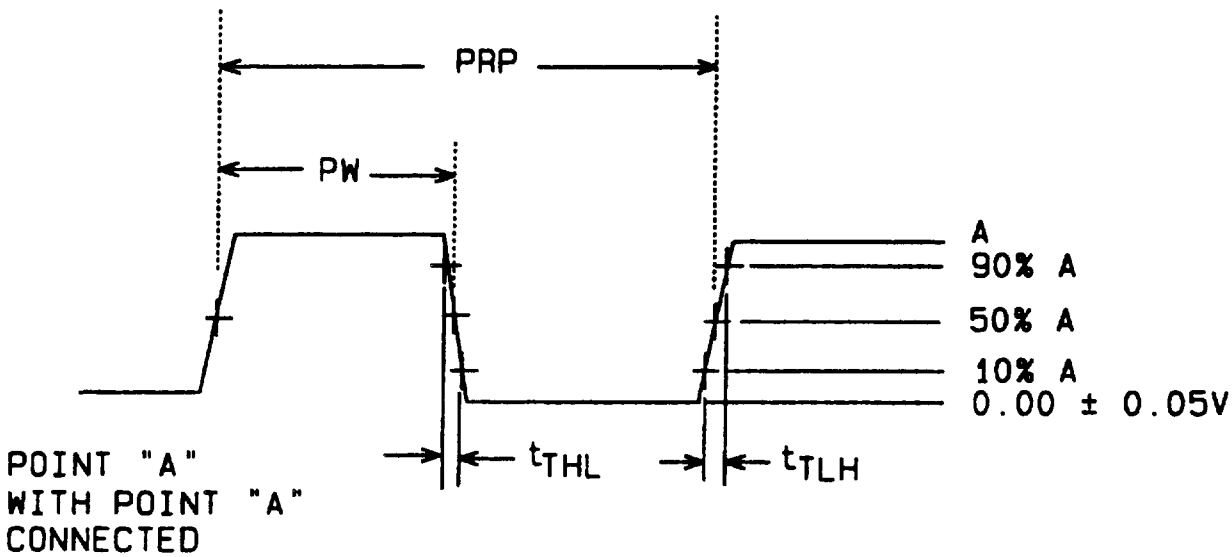


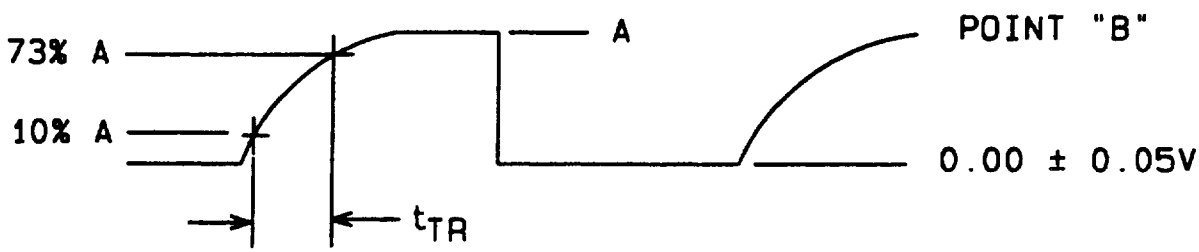
NOTES:

1. R1 = 412 Ohms ± 1 percent, 1/8 W.
2. R2 = 50 Ohms ± 5 percent, 1 W.

* **FIGURE 21.** **Power supply transient rise time test circuit for table XXI.**

Source: https://assist.dla.mil -- Downloaded: 20261O-20T15:22Z
Check the source to verify that this is the current version before use.





NOTE: Input pulse parameters:
A = 0.5V ± 1%
PW = 30 µS ± 10%
PRP = 60 µS ± 10%
$t_{TLH} = t_{THL}$ = 50 nS ± 10%

FIGURE 22. Power supply transient rise time test circuit waveforms for table XXI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-28787/5C	

3. DOCUMENT TITLE
Modules, Standard Electronic, Multiplexer, Digital, Key Code BDL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Technical Point of Contact (TPOC): Mr. Joseph P. Toy (Code 603A) LEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	(1) Commercial TPOC: 812-854-5993	(2) AUTOVON 88-482-5993
c. ADDRESS (Include Zip Code) Commander ATTN: Joseph P. Toy, Code 603A, Bldg. 2917 NAVSURFWARCENDIV, 300 Highway 361 Crane, IN 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:22Z
Check the source to verify that this is the current version before use.